hold for it. This gives

$$(f, f) = \int_{-\pi}^{\pi} |f|^2 \, dx = \sum_{n=-\infty}^{\infty} |c_n|^2 = \pi\left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)\right], \tag{5.53}$$

where we have used Eqs. (5.49).

Let us now summarize the situation. We have shown that the Fourier series converges in the mean to any function $f(x)$ in Hilbert space on $[-\pi, \pi]$. We also know that there exists *some trigonometric sequence* that converges *uniformly* to any *continuous* function $f(x)$ on $[-\pi, \pi]$ for which $f(-\pi) = f(\pi)$. It is plausible therefore that the result for Fourier series, i. e., the particular trigonometric series with the Fourier coefficients [Eqs. (5.48)], can be strengthened. However, it most definitely is not guaranteed that the particular choice of coefficients which guarantees convergence in the mean, also yields uniform convergence. The converse is easily proved, however; namely, if a trigonometric series converges uniformly, then its coefficients are the Fourier coefficients. This may be demonstrated by multiplying [Eq. (5.50)] by cos mx or sin mx and integrating the uniformly convergent series term by term. But establishing the conditions for uniform convergence is the real problem. We begin our investigation of this question, which is the central question in the theory of Fourier series, by proving a theorem.

Theorem 5.4. The convergence of the Fourier series to $f(x)$ is uniform in the closed interval $[-\pi, \pi]$ if $f(x)$ is continuous and its derivative is piecewise continuous in this interval, and $f(-\pi) = f(\pi)$. If, in addition, $f(x + 2\pi) = f(x)$, the convergence will be uniform everywhere.

An example of a continuous function with a piecewise continuous derivative is given in Fig. 5.2. Such functions are often referred to as *piecewise smooth* functions.

Proof. We shall denote the Fourier coefficients of $f'(x)$ by a_n' and b_n'. Integrating by parts, we have

$$\begin{aligned} a_n' &\equiv \frac{1}{\pi}\int_{-\pi}^{\pi} f'(x) \cos nx \, dx = \frac{n}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx \, dx = nb_n , \\ b_n' &\equiv \frac{1}{\pi}\int_{-\pi}^{\pi} f'(x) \sin nx \, dx = -\frac{n}{\pi}\int_{-\pi}^{\pi} f(x) \cos nx \, dx = -n\, a_n , \\ a_0' &= \int_{-\pi}^{\pi} f'(x) \, dx = f(\pi) - f(-\pi) = 0 . \end{aligned} \tag{5.54}$$

We have used the fact that $f(\pi) = f(-\pi)$ in several places; the Fourier coefficients of $f(x)$ are denoted by a_n and b_n as before.

Since $f'(x)$ is piecewise continuous on $[-\pi, \pi]$, and hence square integrable, it must satisfy Bessel's inequality, Eqs. (5.9) and (5.53):

$$(f', f') \geq \pi\left[\frac{a_0'^2}{2} + \sum_{n=1}^{\infty} (a_n'^2 + b_n'^2)\right] = \pi \sum_{n=1}^{\infty} n^2(b_n^2 + a_n^2) , \tag{5.55}$$

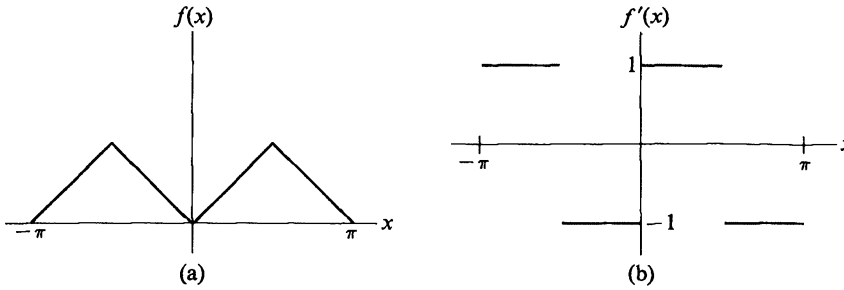

Fig. 5.2. Zig-zag function and its derivative.

where we have used Eqs. (5.54). [In fact, we know that the equality sign holds by the completeness relation, Eq. (5.10)].

The preceding results are all preliminary. We now use the Cauchy criterion to test for the uniform convergence of the Fourier series. Let

$$S_n = \frac{a_0}{2} + \sum_{p=1}^{n} a_p \cos px + \sum_{p=1}^{n} b_p \sin px .$$

By Theorem 5.1, if we can show that $|S_n - S_m| < \epsilon$ for all x in$[-\pi, \pi]$, and for all n, m larger than some $N(\epsilon)$, then we will have established uniform convergence. We have

$$\begin{aligned} |S_n - S_m| &= \left| \sum_{p=m+1}^{n} (a_p \cos px + b_p \sin px) \right| \\ &= \left| \sum_{p=m+1}^{n} \frac{1}{p} (pa_p \cos px + pb_p \sin px) \right| \\ &\leq \sqrt{\sum_{p=m+1}^{n} \left|\frac{1}{p}\right|^2 \sum_{p=m+1}^{n} \left| pa_p \cos px + pb_p \sin px \right|^2} , \end{aligned}$$

where we have used Schwarz's inequality [Eq. (4.8)]. This last expression may be written as

$$\begin{aligned} |S_n - S_m| &\leq \sqrt{\sum_{p=m+1}^{n} \frac{1}{p^2}} \cdot \sqrt{\sum_{p=m+1}^{n} p^2 |a_p \cos px + b_p \sin px|^2} \\ &\leq \sqrt{\sum_{p=m+1}^{n} \frac{1}{p^2}} \cdot \sqrt{\sum_{p=m+1}^{n} p^2 (a_p^2 + b_p^2)} , \end{aligned}$$

since $|a_p \cos px + b_p \sin px|$ can be written as

$$|\sqrt{a_p^2 + b_p^2} \cos (px - \theta)| \leq |\sqrt{a_p^2 + b_p^2}| ,$$

where $\theta = \tan^{-1} b_p/a_p$. Therefore

$$\left| S_n - S_m \right| \leq \sqrt{\sum_{p=m+1}^{n} \frac{1}{p^2}} \cdot \sqrt{\frac{1}{\pi} \int_{-\pi}^{\pi} |f'(x)|^2 \, dx} ,$$

where the term $\sqrt{\sum_{p=m+1}^{n} p^2(a_p^2 + b_p^2)}$ has been replaced by the equal or larger term

$$\sqrt{\sum_{p=1}^{\infty} p^2(a_p^2 + b_p^2)} \leq \sqrt{(1/\pi)\int_{-\pi}^{\pi} |f'|^2\, dx}\,, \tag{5.56}$$

from Eq. (5.55). Now the series $\sum_{p=1}^{\infty}(1/p^2)$ converges (it is, in fact, the Riemann zeta function evaluated at 2 and converges to $\pi^2/6$ as we shall show in Section 6.9). If we let

$$M = \sqrt{(1/\pi)\int_{-\pi}^{\pi} |f'(x)|^2\, dx}\,,$$

then we know that M is finite, because $f'(x)$ is piecewise continuous and always finite.

The series $\sum_{p=1}^{\infty}(1/p^2)$, converges, so it must satisfy the Cauchy criterion. Then for any positive number, which we take to be ϵ^2/M^2, there exists an integer N such that $\sum_{p=m+1}^{n}(1/p^2) < \epsilon^2/M^2$ when $n, m > N$. Taking the n and m that appear in $|S_n - S_m|$ larger than this N, we then have

$$|S_n - S_m| < (\epsilon^2/M^2)^{1/2} M = \epsilon\,,$$

and the proof of uniform convergence is complete. We know that the series converges *to* the function $f(x)$ because its convergence in the mean to $f(x)$ has already been established.

Note that the completeness relation could have been used earlier in this proof; there would then be an equals sign where we have an inequality in Eq. (5.56). It *is* valid to use the completeness relation because completeness has been proved already. But in order to demonstrate that prior knowledge of completeness is not required to prove uniform convergence, we have used Bessel's inequality instead. The piecewise continuity of $f'(x)$ implies that M is finite, and this is all that is needed to prove uniform convergence. However, to show that the uniform convergence is *to* $f(x)$ (since Cauchy's criterion makes no mention of the limit function) the convergence in the mean to $f(x)$ (completeness) was used at the very end of the proof. It is possible to prove independently that the uniform convergence is to $f(x)$, and then this result leads to an independent proof of completeness.

This theorem may be extended to deal with the piecewise continuous functions, such as step functions, which may have a finite number of finite discontinuities in the interval $[-\pi, \pi]$.

Theorem 5.5. If $f(x)$ is piecewise continuous in $[-\pi, \pi]$, and has a piecewise continuous derivative there, then its Fourier series converges uniformly to $f(x)$ in every closed subinterval of $[-\pi, \pi]$ in which $f(x)$ is continuous. At points of discontinuity of $f(x)$, its Fourier series converges to the arithmetic mean of the left- and right-hand limits of the function. If $f(x + 2\pi) = f(x)$, then these statements are true everywhere on the real line.

It is interesting that although we know that there exists some *trigonometric* sequence that converges uniformly to every continuous function on a closed interval, continuity of $f(x)$ alone is not sufficient to prove the covergence of its *Fourier* series. There exist continuous functions which are not differentiable, and their *Fourier* series need not converge. The additional condition of continuity of $f'(x)$, however, ensures convergence. Also, the assumption that the intervals be closed is vital to proving *uniform* convergence. The convergence of the Fourier series for piecewise continuous functions—such as step functions—will certainly not be uniform in open intervals (a, b), where one of the endpoints is a point of discontinuity. In fact, it may be shown that just before the Fourier series passes over the discontinuity, it differs from the function by a finite amount. This overshooting effect is called the Gibbs phenomenon (see Problem 4c).

It is primarily the smoothness of the function that determines the size of the Fourier coefficients—the smoother the function, the more rapidly these coefficients decrease and the more rapid is the convergence. The theory of Fourier series is largely the study of the interplay between assumptions about smoothness and conclusions about convergence.

Fejer has constructed a theory of Fourier series based on a special type of series summation (called Cesaro summation) in which one considers the sequence of *arithmetic means* of the partial sums. If a series is convergent in the usual sense, then it is Cesaro summable to the same value. But many divergent series are Cesaro summable; thus Cesaro summability is a natural generalization since it preserves the usual summation as a special case. It was shown by Fejer that the Fourier series of any continuous function $f(x)$ is uniformly Cesaro summable to $f(x)$ (see Rudin or Apostol). No assumptions on smoothness are required. This result is to be contrasted to the fact that there are continuous functions whose Fourier series are divergent at a point.

So far we have restricted our attention to functions defined on the interval $[-\pi, \pi]$. If these functions are periodic of period 2π, the expansions are as good everywhere as within the interval $[-\pi, \pi]$. It is a simple matter to generalize these results to the interval $[-l, l]$, and functions of period $2l$. But the requirement of periodicity cannot be removed if the series is to converge outside the basic interval.

Clearly the set of functions

$$\left\{\frac{e^{in\pi x/l}}{(2l)^{1/2}}\right\}, \qquad n = 0, \pm 1, \pm 2, \cdots$$

is a complete orthonormal set on the interval $[-l, l]$. All our results go through for this set of functions exactly as before. For the complex exponential series, we obtain

$$\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty} c_n \frac{e^{in\pi x/l}}{(2l)^{1/2}}, \\ c_n &= \frac{1}{(2l)^{1/2}} \int_{-l}^{l} f(x) e^{-in\pi x/l}\, dx . \end{aligned} \tag{5.57}$$

Similarly, for the trigonometric series,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{l} x + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{l} x ,$$

$$a_n = \frac{1}{l} \int_{-l}^{l} f(x) \cos \frac{n\pi}{l} x \, dx \qquad \text{for } n = 0, 1, \cdots, \tag{5.58}$$

$$b_n = \frac{1}{l} \int_{-l}^{l} f(x) \sin \frac{n\pi}{l} x \, dx \qquad \text{for } n = 1, 2, \cdots$$

These formulas hold in the interval $[-l, l]$ under the conditions of the theorems, and will hold outside this interval only if $f(x + 2l) = f(x)$.

For any l (including $l = \pi$), if $f(x)$ is an even function, then $b_n = 0$ for all n and we have a Fourier cosine series; and if $f(x)$ is an odd function, $a_n = 0$ for each n and we have a Fourier sine series.

We close this section with a very simple, but very important observation. It will be recalled that the Legendre polynomials, which we have shown to be a complete orthonormal set, satisfy a second-order differential equation (Eq. 5.44). In this section we have been discussing the more familiar trigonometric functions, and have found that they also are a complete orthonormal set of functions. As we study other such sets of functions we shall always make note of the fact that these functions are the solutions of certain differential equations. Our treatment will not be derived from these differential equations, but rather will culminate in them, deriving them as a by-product of other considerations. Finally, by way of a summary, we shall, in Section 5.11, view all the special polynomials in their capacity as the solutions of second-order differential equations.

The trigonometric functions will, of course, not be included in this summary, since they are not polynomials. For completeness sake, and to help motivate the next section on Fourier integrals, we state the well-known differential equation satisfied by the sine and cosine functions:

$$\frac{d^2}{dx^2} u + \omega^2 u = 0 . \tag{5.59}$$

This second-order differential equation has the two linearly independent solutions

$$u_1 = \sin \omega x, \qquad u_2 = \cos \omega x . \tag{5.60}$$

If $\omega^2 = n^2$, where n is an integer, these solutions give the Fourier functions as n runs from 0 to ∞. It is instructive to prove the orthogonality of these functions from the form of the differential equation as was done for the Legendre polynomials.

5.7. FOURIER INTEGRALS

The restriction of the validity of Fourier series expansions to the basic interval $[-l, l]$ unless the expanded function is periodic is annoying because many functions are not periodic. However, a way around this difficulty is suggested

by the generalization we made in the last section: If we want to expand a nonperiodic function over any specified finite range, all we have to do is expand the function in an interval $[-l, l]$ large enough to contain the specified range. But what if we want an expansion for a nonperiodic function that is valid everywhere, over the whole real line? Throwing all considerations of rigor aside, we shall here derive heuristically a plausible answer to this question. (When a physicist speaks of a "heuristic" derivation, he is usually apologizing in advance for a sloppy, nonrigorous one; actually, the word means "serving to help toward discovery.") We shall take up these matters again, in a more rigorous fashion, in Chapter 9 (corollary to Theorem 9.11).

It is reasonable to try to get around the periodicity requirement by letting $l \to \infty$ in Eqs. (5.57). To carry this out, set $(\pi/l)^{1/2}x = y$ and $n(\pi/l)^{1/2} = k_n$, so $(n\pi/l)\,x = k_n y$ and $\Delta k_n \equiv k_{n+1} - k_n = (\pi/l)^{1/2}$, and $1/(2l)^{1/2} = \Delta k_n/(2\pi)^{1/2}$. Then, replacing c_n by g_{k_n}, Eqs. (5.57) become

$$f(y) = \frac{1}{(2\pi)^{1/2}} \sum_{k_n=-\infty}^{\infty} g_{k_n} e^{ik_n y}\, \Delta k_n ,$$

where

$$g_{k_n} = \frac{1}{(2\pi)^{1/2}} \int_{-\sqrt{\pi}l}^{\sqrt{\pi}l} f(y) e^{-ik_n y}\, dy ,$$

and k_n changes by steps of $\Delta k_n = (\pi/l)^{1/2}$, corresponding to steps of $\Delta n = 1$ in the original sum. Now let $l \to \infty$. As $l \to \infty$, $\Delta k_n \to 0$ and k_n takes on all real values. Therefore the sum over k_n becomes an integral over a continuous variable (which we denote by k), and we have, replacing y by x,

$$\begin{aligned} f(x) &= \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} g(k) e^{ikx}\, dk , \\ g(k) &= \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} f(x) e^{-ikx}\, dx . \end{aligned} \tag{5.61}$$

The function $g(k)$ is called the *Fourier transform* of $f(x)$, and vice versa. Two functions which satisfy Eqs. (5.61) are called a Fourier transform pair. This pair may be regarded as the statement and solution of an integral equation of a particular type.

By an easy extension of these results, we get the three-dimensional Fourier integral:

$$\begin{aligned} F(\mathbf{r}) &= \frac{1}{(2\pi)^{3/2}} \int_{-\infty}^{\infty} G(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}}\, d\mathbf{k} , \\ G(\mathbf{k}) &= \frac{1}{(2\pi)^{3/2}} \int_{-\infty}^{\infty} F(\mathbf{r}) e^{-i\mathbf{k}\cdot\mathbf{r}}\, d\mathbf{r} . \end{aligned} \tag{5.62}$$

Here $\mathbf{r}$ stands for x, y, z, and $d\mathbf{r} = dx\,dy\,dz$; similarly, $\mathbf{k}$ stands for k_x, k_y, k_z, and $d\mathbf{k} = dk_x\,dk_y\,dk_z$. Thus $\mathbf{k}\cdot\mathbf{r} = k_x x + k_y y + k_z z$. The integrals are threefold. We shall use this three-dimensional Fourier transform pair to solve some of the partial differential equations of physics in Chapter 7.

Equation (5.61) may be proved rigorously if one assumes that $f(x)$ is piecewise smooth and that $\int_{-\infty}^{\infty}|f(x)|\,dx$ exists (see Courant-Hilbert), but we shall not go into these issues here (see Section 9.6). Instead, we shall focus on the uses of the Fourier transform in physics.

Example. An interesting result may be obtained if we combine Eqs. (5.61) into the single equation

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} f(x')e^{-ikx'}\,dx'\right]e^{ikx}\,dk\,,$$

and then interchange the orders of integration to get

$$f(x) = \int_{-\infty}^{\infty} f(x')\left[\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{ik(x-x')}\,dk\right]dx'\,.$$

The expression in brackets within the integration over x' must be a δ-function if this equation is to hold. Therefore we have an integral representation of the δ-function:

$$\delta(x - x') = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{ik(x-x')}\,dk\,. \tag{5.63}$$

This equation is to be interpreted as meaning that the expression on either side of it will have the same effect under an integral sign. This "derivation" shows clearly that the δ-functions (as naively conceived and used) originate as a consequence of the illegitimate business of interchanging orders of integration; the original crime must be paid for by the introduction of the extraordinary new mathematical objects called δ-functions. Their symbolic use is so simple, however, that one is apt to get the impression that the crime paid. In a sense it did: violations of the existing rules in mathematics and physics are always required to reach the new, which then turns out to hide the old as a special case.

The δ-function in three dimensions is

$$\delta^3(\mathbf{r} - \mathbf{r}_0) \equiv \delta(x - x_0)\delta(y - y_0)\delta(z - z_0) = \frac{1}{(2\pi)^3}\int_{-\infty}^{\infty} e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}_0)}\,d\mathbf{k}\,. \tag{5.64}$$

The integral representation of the δ-function can be used to prove the completeness relation (sometimes called Parseval's theorem) for the Fourier integral. Thus

$$\begin{aligned}\int_{-\infty}^{\infty}|f(x)|^2\,dx &= \int_{-\infty}^{\infty} dx\,\frac{1}{(2\pi)^{1/2}}\int_{-\infty}^{\infty} g^*(k)e^{-ikx}\,dk\,\frac{1}{(2\pi)^{1/2}}\int_{-\infty}^{\infty} g(k')e^{ik'x}\,dk' \\ &= \int_{-\infty}^{\infty} dk\,g^*(k)\int_{-\infty}^{\infty} dk'\,g(k')\left[\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i(k'-k)x}\,dx\right] \\ &= \int_{-\infty}^{\infty} dk\,g^*(k)\int_{-\infty}^{\infty} dk'\,g(k')\delta(k' - k) \\ &= \int_{-\infty}^{\infty} g^*(k)\,g(k)\,dk = \int_{-\infty}^{\infty}|g(k)|^2\,dk\,.\end{aligned}$$

If x is the time t, and k is the frequency υ, then Parseval's theorem has a simple physical interpretation. It reads

$$\int_{-\infty}^{\infty} |f(t)|^2 \, dt = \int_{-\infty}^{\infty} |g(\upsilon)|^2 \, d\upsilon \, .$$

If, for example, $f(t)$ is a radiated electric field, then $|f(t)|^2$ is proportional to the total radiated power, and the integral is a measure of the total radiated energy. On the right-hand side is the integral over all frequencies of the spectrum amplitude squared, $|g(\upsilon)|^2$, which is proportional to the energy radiated per unit frequency interval. Thus Parseval's theorem expresses the conservation of energy.

Before turning to some applications, we state and prove the convolution theorem for Fourier transforms. We shall use it in Chapter 7. Let $f_1(x)$ and $f_2(x)$ be two functions whose Fourier transforms, $g_1(k)$ and $g_2(k)$, are given by Eq. (5.61).

The quantity

$$F(x) \equiv \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} f_1(t) f_2(x - t) \, dt$$

is called the *convolution* of the functions f_1 and f_2. The convolution theorem states that

$$F(x) = \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} g_1(k) g_2(k) e^{ikx} \, dk \, . \tag{5.65}$$

By taking the Fourier transform of this last equation, and thus solving for the product $g_1(k)\, g_2(k)$, we see that the convolution theorem may also be expressed as follows: The Fourier transforms of the convolution of two functions is the product of the Fourier transforms of these two functions. Denoting the Fourier transform of $F(x)$ by $G(k)$ this may be written

$$G(k) = g_1(k)\, g_2(k) \, .$$

The proof follows immediately from the properties of the Fourier-transform pair. Thus

$$\begin{aligned} F(x) &= \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} f_1(t) f_2(x - t) \, dt \\ &= \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} f_1(t) \left[\frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} g_2(k) e^{ik(x-t)} \, dk \right] dt \\ &= \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} \left[\frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} f_1(t) e^{-ikt} \, dt \right] g_2(k) e^{ikx} \, dk \\ &= \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} g_1(k) g_2(k) e^{ikx} \, dk \, , \end{aligned}$$

which establishes the convolution theorem.

We shall now examine, in some detail, an example of the use of Fourier transforms, and incidentally, the calculation of some specific transforms.

We consider an undamped, one-dimensional harmonic oscillator acted upon by a time-varying but spatially uniform (no x-dependence) external force $F(t)$. This might be, for example, a spring hanging from a board which is jerked upward. We want to compute the energy transferred to the oscillator by such a force. Throughout this example, we shall limit ourselves to force functions $F(t)$ for which the Fourier transform exists.

The equation of motion is

$$\ddot{x} + \omega^2 x = (1/m)F(t) , \tag{5.66}$$

where m is the mass and ω is the natural frequency of the oscillator. We may rewrite this as

$$\dot{z} - i\omega z = (1/m)F(t) , \tag{5.67}$$

where $z = \dot{x} + i\omega x$. The energy of the oscillator at any time t is given by

$$E(t) = m\dot{x}^2/2 + m\omega^2 x^2/2 = (m/2)|z(t)|^2 .$$

Let us now assume that the oscillator is originally at rest ($x = \dot{x} = z = 0$ for $t \leq T_1$). Since the oscillator initially has zero energy, the energy ΔE transferred to the oscillator by $F(t)$ is $E(\infty)$. We now compute this quantity; we shall find that it depends in a simple way on the Fourier transform of $F(t)$.

Multiplying both sides of Eq. (5.67) by $\exp(-i\omega t)$ and integrating from T_1 to T_2 (arbitrary times), we obtain

$$\int_{T_1}^{T_2} (\dot{z} - i\omega z)\, e^{-i\omega t}\, dt = (1/m) \int_{T_1}^{T_2} F(t) e^{-i\omega t}\, dt .$$

Integration by parts of the term on the left gives

$$z(T_2)e^{-i\omega T_2} = (1/m) \int_{T_1}^{T_2} F(t) e^{-i\omega t}\, dt ,$$

where we have used the boundary condition that z vanishes for $t \leq T_1$. Since $F(t)$ vanishes for $t \leq T_1$ we may extend the lower limit of the integration to $-\infty$. The energy transfer is then given by

$$\begin{aligned} \Delta E &= \lim_{T_2 \to \infty} \frac{m}{2} |z(T_2)|^2 \\ &= \lim_{T_2 \to \infty} \frac{m}{2} \left| e^{i\omega T_2} \frac{1}{m} \int_{-\infty}^{T_2} F(t) e^{-i\omega t}\, dt \right|^2 \\ &= \frac{1}{2m} \left| \int_{-\infty}^{\infty} F(t) e^{-i\omega t}\, dt \right|^2 \\ &\equiv \frac{1}{2m} |f(\omega)|^2 , \end{aligned} \tag{5.68}$$

where $f(\omega)$ is the Fourier transform of $F(t)$ evaluated at the natural frequency of the oscillator:

$$f(\omega) \equiv \int_{-\infty}^{\infty} F(t) e^{-i\omega t}\, dt .$$

Thus $F(t)$ is given in terms of $f(\omega)$ by

$$F(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(\omega)e^{i\omega t}\,d\omega\,.$$

We have consolidated the factors of $(2\pi)^{-1/2}$ in one member of this transform pair, which therefore differs slightly in appearance from the symmetric form of Eqs. (5.61). Thus the energy transfer is essentially the absolute value squared of the Fourier component of the force function whose frequency is the natural frequency of the oscillator. The energy transfer can only occur at the resonant frequency ω, because only the ω-component of the Fourier transform of $F(t)$ enters into the formula for ΔE.

To solve the energy transfer problem for a specific force function, we must find its Fourier transform. We develop here some formulas which facilitate the determination of Fourier transforms. Let

$$F_n(t) \equiv \frac{d^{n-1}F(t)}{dt^{n-1}}\,, \qquad F_1(t) \equiv F(t)\,,$$

$$f_n(\omega) = \int_{-\infty}^{\infty} F_n(t)e^{-i\omega t}\,dt\,.$$

An integration by parts gives

$$f_n(\omega) = \left(\frac{1}{i\omega}\right)f_{n+1}(\omega) - \frac{1}{i\omega}\,F_n(t)\,e^{-i\omega t}\Big]_{-\infty}^{\infty}\,.$$

This formula will be useless unless $F_n(-\infty) = F_n(+\infty) = 0$. We therefore consider only such cases. Then, by repeated integration by parts (i.e., iteration of the above formula), we can express $f_1(\omega)$, the Fourier transform needed for this problem, in terms of the Fourier transform of some higher derivative of $F(t)$. This may facilitate the determination of $f_1(\omega)$ if the Fourier transform of a higher derivative is known. We obtain for $f_1(\omega) \equiv f(\omega)$,

$$f(\omega) = \left(\frac{1}{i\omega}\right)^{n-1} f_n(\omega)\,. \tag{5.69}$$

The index n is arbitrary, but in using the formula one naturally chooses the smallest n for which $f_n(\omega)$ is known.

We shall now compute the Fourier transforms of several force functions, and the energy transferred to an oscillator subjected to these forces.

Example 1. *Impulsive force*: $F(t) = P_0\delta(t)$. We obtain

$$\Delta E = \frac{1}{2m}\left|\int_{-\infty}^{\infty} P_0\delta(t)e^{-i\omega t}\,dt\right|^2 = P_0^2/2m\,.$$

This result may also be obtained by simply dropping the term $\omega^2 x$ in the equation of motion (5.66), the so-called impulse approximation.

Example 2. *Gaussian force*: $F(t) = (P_0/\sqrt{\pi}\tau)e^{-t^2/\tau^2}$ (Fig. 5.3). The width of the Gaussian pulse at half-maximum is $2\sqrt{\ln 2}\tau$. The Fourier transform may

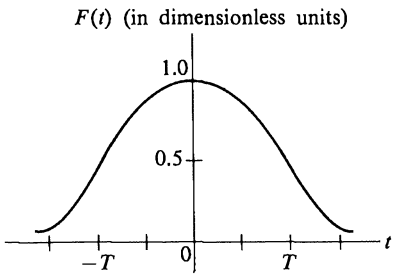

Fig. 5.3.

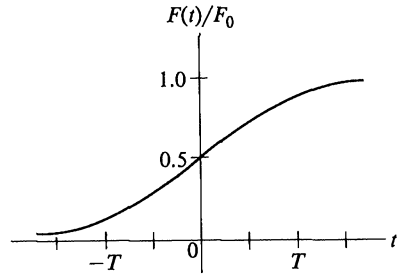

Fig. 5.4.

be computed directly by completing the square in the exponent:

$$\begin{aligned} f(\omega) &= \frac{P_0}{\sqrt{\pi}\tau}\int_{-\infty}^{\infty} e^{-t^2/\tau^2}e^{-i\omega t}\,dt \\ &= \frac{P_0}{\sqrt{\pi}\tau}\,e^{-\omega^2\tau^2/4}\int_{-\infty}^{\infty} e^{-(t+i\omega\tau^2/2)^2/\tau^2}\,dt \\ &= \frac{P_0}{\pi^{1/2}}\,e^{-\omega^2\tau^2/4}\int_{-\infty}^{\infty} e^{-x^2}\,dx = P_0 e^{-\omega^2\tau^2/4}\ . \end{aligned}$$

By using techniques that will be developed in Section 6.9, we can show that this formal procedure for evaluating the integral is in fact valid. Note that the Fourier transform of a Gaussian is a Gaussian. For the energy transferred to the oscillator, we find that

$$\Delta E = (P_0^2/2m)e^{-\omega^2\tau^2/2}\ .$$

In the "sudden" limit ($\tau << \omega^{-1}$), $\Delta E = (P_0^2/2m)$, as in Example 1. This is to be expected because in the limit $\tau \to 0$, the Gaussian force function becomes a δ-function (see Example 2, Section 5.3). In the adiabatic limit ($\tau >> \omega^{-1}$), there is no energy transferred to the oscillator which also makes sense physically.

Example 3. *Probability integral*:

$$F(t) = (F_0/\sqrt{\pi}\tau)\int_{-\infty}^{t} e^{-t'^2/\tau^2}\,dt'$$

(Fig. 5.4). In this case the Fourier integral does not really exist because the function assumes a nonzero value (F_0) asymptotically as $t \to \infty$. We may still use Eqs. (5.68) and (5.69) to calculate the energy transfer, imagining that the constant force is turned off adiabatically at very large times. This will not affect the net energy transfer, and will bring about convergence of the Fourier integral. We have already determined the Fourier transform of

$$dF(t)/dt \equiv F_2(t) = (F_0/\sqrt{\pi}\tau)e^{-t^2/\tau^2}$$

in Example 2. Using Eq. (5.69) with $n = 2$ and Eq. (5.68), we obtain for the

energy transfer

$$\Delta E = \frac{1}{2m}\left| f(\omega) \right|^2 = \frac{1}{2m}\left|\left(\frac{1}{i\omega}\right)f_2(\omega)\right|^2 = \frac{1}{2m}(F_0/\omega)^2 e^{-\omega^2\tau^2/2} .$$

Again, in the adiabatic limit, $\Delta E \to 0$; in the sudden limit, $\Delta E \to 1/2m\,(F_0/\omega)^2$. In the latter case, the mass, because of its inertia, suddenly finds itself displaced a distance $l_0 = (F_0/m\omega^2)$ from equilibrium and hence acquires the energy $(\frac{1}{2}m\omega^2 l_0^2)$ corresponding to the full displacement. It has no time to follow the shifting equilibrium position in the sudden limit. This result for the sudden limit should agree with the limiting behavior of the energy transfer for the force functions of Problems 7 and 8.

5.8 SPHERICAL HARMONICS AND ASSOCIATED LEGENDRE FUNCTIONS

We have derived the completeness of the Legendre polynomials from Weierstrass's theorem in one variable. From the two-variable generalization of Weierstrass's theorem, we proved the completeness of the trigonometric functions. We now derive the completeness of the spherical harmonics from Weierstrass's for three variables. It tells us that a function F of x, y, z (that is, $\mathbf{r}$) can be approximated uniformly by a sequence of partial sums as follows:

$$F(\mathbf{r}) = \lim_{M\to\infty} F_M(\mathbf{r}) = \lim_{M\to\infty} \sum_{m,\,n,\,p=0}^{M} c_{mnp}^{(M)} x_1^m x_2^n x_3^p . \tag{5.70}$$

We may also express $F_M(\mathbf{r})$ in terms of the three variables:

$$z_1 \equiv x_1 + ix_2 = r\sin\theta e^{i\phi} ,$$
$$z_2 \equiv x_1 - ix_2 = r\sin\theta e^{-i\phi} ,$$
$$z_3 \equiv x_3 = r\cos\theta ,$$

which are linear combinations of x_1, x_2, x_3. Thus

$$\begin{aligned} F_M(\mathbf{r}) &= \sum_{\alpha,\,\beta,\,\gamma=0}^{M} A_{\alpha\beta\gamma}^{(M)} z_1^\alpha z_2^\beta z_3^\gamma \\ &= \sum_{l=0}^{3M} r^l \sum_{\substack{\alpha,\,\beta,\,\gamma=0 \\ (\alpha+\beta+\gamma=l)}}^{M} A_{\alpha\beta\gamma}^{(M)} e^{i(\alpha-\beta)\phi} \sin^{(\alpha+\beta)}\theta \cos^\gamma\theta . \end{aligned} \tag{5.71}$$

In the last expression we have converted the unrestricted sum over all values of α, β, γ to a sum over just those combinations of α, β, γ that sum to the integer l. But then we sum over all possible restrictions (all l), which in effect removes the restriction on α, β, γ and gives the same results as the original unrestricted sum.

We now restrict $F(\mathbf{r})$ to the unit sphere by requiring that $r = 1$; also, we set $(\alpha - \beta) = m$. We shall need the following facts: Since α, β, and γ are

all positive or zero, we have

$$\alpha + \beta = m + 2\beta \geq 0 .$$

Also,

$$\alpha + \beta \geq |\alpha - \beta| = |m| \Longrightarrow \alpha + \beta - |m| = m - |m| + 2\beta \geq 0 .$$

Furthermore, $(\alpha + \beta - |m|)$ is always even because 2β is even and $m - |m| = 0$ if $m \geq 0$ and $m - |m| = -2m$ if $m < 0$. We now rewrite Eq. (5.71) in the form

$$F_M(\theta, \phi) = \sum_{l=0}^{3M} \sum_{\substack{\alpha, \beta, \gamma=0 \\ (\alpha+\beta+\gamma=l)}}^{M} A_{\alpha\beta\gamma}^{(M)} e^{im\phi} \sin^{(\alpha+\beta-|m|)} \theta \cos^{\gamma} \theta \sin^{|m|} \theta .$$

By a trigonometric identity,

$$\sin^{(\alpha+\beta-|m|)}\theta \cos^{\gamma} \theta = (1 - \cos^2 \theta)^{(\alpha+\beta-|m|)/2} \cos^{\gamma} \theta ,$$

which is a polynomial in $\cos\theta$ of maximum degree $\alpha + \beta + \gamma - |m| = l - |m|$, since $(\alpha + \beta - |m|)$ is even. Denoting this polynomial by $f_{lm}(\cos\theta)$, we get

$$F_M(\theta, \phi) = \sum_{l=0}^{3M} \sum_{m} B_{lm}^{(M)} e^{im\phi} \sin^{|m|} \theta f_{lm}(\cos\theta) .$$

The range of the summation over m is still to be specified. Since $m = \alpha - \beta$, $m \leq l$; also, since we sum only over nonnegative α, β, γ, $\alpha + \beta + \gamma - |m| = l - |m| \geq 0$. Therefore, in changing to a sum over m, we must only sum over those m's such that $|m| \leq l$. Thus the last equation becomes

$$F_M(\theta, \phi) = \sum_{l=0}^{3M} \sum_{m=-l}^{m=l} B_{lm}^{(M)} e^{im\phi} \sin^{|m|} \theta f_{lm}(\cos\theta) . \tag{5.72}$$

Therefore the sequence of functions

$$\overline{Y}_{lm}(\theta, \phi) \equiv e^{im\phi} \sin^{|m|} \theta f_{lm}(\cos\theta) ,$$

where $f_{lm}(\cos\theta)$ is a polynomial in $\cos\theta$ of degree $(l - |m|)$, provides a uniform approximation to any continuous function defined on the unit sphere—that is, over the range $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$. If we can now construct from this set of functions an *orthonormal* set of functions, then by reasoning exactly analogous to that applied to the Legendre polynomials and the trigonometric functions, we will know that this set of functions is complete. This is just another instance of the use of our basic result on completeness: whenever we can uniformly approximate an arbitrary function in Hilbert space by a sequence of partial sums which are themselves linear combinations of a set of orthonormal functions, then this set is a complete orthonormal set in Hilbert space.

Thus let us look for functions Y_{lm} satisfying

$$\int_{\Omega} Y_{l'm'}^* Y_{lm} \, d\Omega \equiv \int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta \, d\theta \, Y_{l'm'}^* Y_{lm} = \delta_{ll'}\delta_{mm'} . \tag{5.73}$$

This orthonormality condition uniquely determines the functions Y_{lm} up to a phase factor, exactly as in the case of the normalized Legendre polynomials, $\bar{P}_l$. Thus there can only be one complete set of orthonormal functions defined on the unit sphere. We shall calculate a few of the Y_{lm} explicitly.

If $l = m = 0$, we obtain $Y_{00} = (1/4\pi)^{1/2}$. If $l = 1$, then m may equal -1, 0, or $+1$. Recalling that $f_{lm}(\cos\theta)$ is a polynomial in $\cos\theta$ of degree $l - |m|$, we obtain

$$Y_{10} = a_1 \cos\theta + a_2 ,$$
$$Y_{11} = a_3 e^{i\phi} \sin\theta ,$$
$$Y_{1,-1} = a_4 e^{-i\phi} \sin\theta .$$

The constants a_1, a_2, a_3, a_4 are determined by imposing orthonormality. Thus

$$0 = \delta_{01}\delta_{00} = \int Y_{00}^* Y_{10}\, d\Omega$$
$$= \frac{1}{(4\pi)^{1/2}} \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \sin\theta (a_1 \cos\theta + a_2) \Longrightarrow a_2 = 0 ,$$
$$1 = \delta_{10}\delta_{10} = \int |Y_{10}|^2\, d\Omega \Longrightarrow a_1 = \left(\frac{3}{4\pi}\right)^{1/2} .$$

Similarly, $a_3 = -a_4 = -(3/8\pi)^{1/2}$. We choose the minus sign to be consistent with the convention to be adopted later, in Eq. (5.75). Therefore the first few members of the complete orthonormal set of functions on the unit sphere are

$$Y_{00} = \left(\frac{1}{4\pi}\right)^{1/2}, \qquad Y_{11} = -\left(\frac{3}{8\pi}\right)^{1/2} e^{i\phi} \sin\theta ,$$
$$Y_{10} = \left(\frac{3}{4\pi}\right)^{1/2} \cos\theta , \qquad Y_{1,-1} = \left(\frac{3}{8\pi}\right)^{1/2} e^{-i\phi} \sin\theta . \tag{5.74}$$

Note that for a given l there are $2l + 1$ functions Y_{lm}. The functions Y_{lm} are called *spherical harmonics*. They are a sort of two-dimensional generalization and combination of the Fourier functions and the Legendre polynomials, since they do for the spherical *surface* what the Fourier functions and the Legendre polynomials do for their respective linear intervals.

We give next a general formula for the Y_{lm}. Since we know that the Y_{lm} are unique, the formula is true if we establish orthonormality—just as in the case of the Legendre polynomials. The formula is

$$Y_{lm}(\theta, \phi) = (-1)^m \left[\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}\right]^{1/2} P_l^m(\cos\theta) e^{im\phi} , \qquad m \geq 0 , \tag{5.75}$$
$$Y_{l,-m} = (-1)^m Y_{lm}^* , \qquad m \geq 0 ,$$

where

$$P_l^m(x) = (1 - x^2)^{m/2} \frac{d^m}{dx^m} P_l(x) , \qquad m \geq 0 , \tag{5.76}$$

are called the *associated Legendre functions.* Thus the functions

$$Y_{l0}(\theta, \phi) = \left(\frac{1}{2\pi}\right)^{1/2}\left(\frac{2l+1}{2}\right)^{1/2} P_l(\cos\theta)$$

are an orthonormal set of functions on the unit sphere.

We shall first show that if u is a solution of Legendre's differential equation, that is,

$$(1 - x^2)u'' - 2xu' + l(l+1)u = 0\,, \tag{5.77}$$

then $v \equiv (1 - x^2)^{m/2}(d^m/dx^m)u$, for integral $m \geq 0$, is a solution of the equation

$$(1 - x^2)v'' - 2xv' + [l(l+1) - m^2/(1 - x^2)]v = 0\,, \tag{5.78}$$

known as Legendre's associated equation. Let $d^m u/dx^m \equiv d^m u \equiv w$. Then

$$v = (1 - x^2)^{m/2}w\,, \qquad v' = (1 - x^2)^{m/2}w' - mx(1 - x^2)^{(m/2)-1}w\,,$$

$$v'' = (1 - x^2)^{m/2}w'' - 2mx(1 - x^2)^{(m/2)-1}w' - m(1 - x^2)^{(m/2)-1}w + mx^2(m-2)(1 - x^2)^{(m/2)-2}w\,.$$

Substituting into (5.78) and replacing w by $d^m u$, the left-hand side of Eq. (5.78) becomes

$$(1 - x^2)d^{m+2}u - 2(m+1)x\,d^{m+1}u + [l(l+1) - m(m+1)]d^m u\,. \tag{5.79}$$

We shall now show that this expression can be written in the form

$$d^m[(1 - x^2)u'' - 2xu' + l(l+1)u]\,, \tag{5.80}$$

which vanishes, by comparison with Eq. (5.77). This completes the proof that $v = (1 - x^2)^{m/2}(d^m u/dx^m)$ is a solution of Legendre's associated equation. Using Leibnitz's formula [Eq. (5.42)], we obtain

$$d^m[(1 - x^2)u''] = \sum_{n=0}^{m}\binom{m}{n} d^n(1 - x^2)d^{m-n+2}u$$
$$= (1 - x^2)d^{m+2}u - 2xm\,d^{m+1}u - m(m-1)d^m u\,,$$

and also

$$\frac{d^m}{dx^m}[-2xu'] = -2xd^{m+1}u - 2md^m u\,.$$

Combining the last two lines and the term $l(l+1)d^m u$ from Eq. (5.80) establishes the equivalence of Eqs. (5.79) and (5.80) and completes the proof that the functions

$$P_l^m(x) = (1 - x^2)^{m/2}\frac{d^m}{dx^m}P_l(x) = \frac{1}{2^l l!}(1 - x^2)^{m/2}\frac{d^{l+m}}{dx^{l+m}}(x^2 - 1)^l \tag{5.81}$$

are solutions of Legendre's associated equation (5.78). We note that $P_l^0(x) = P_l(x)$.

Using the same procedure that led to the orthonormality relations for the Legendre polynomials, we can show that

$$\int_{-1}^{1} P_l^m(x) P_{l'}^m \, dx = \frac{(l+m)!}{(l-m)!} \frac{2}{2l+1} \delta_{ll'} . \tag{5.82}$$

Once this result has been established, the normalization constant of the Y_{lm} (Eq. 5.75) follows immediately, as the integration over ϕ merely produces a factor of 2π. There is, of course, a free choice of phase factor; ours is a common choice in the physics literature. However, one must be careful, because different authors choose different phase factors for the spherical harmonics. The $P_l^m(x)$ are *not* another orthonormal set of polynomials on $[-1, 1]$, challenging the uniqueness of the Legendre polynomials, for the simple reason that they are not polynomials at all! Equation (5.81) shows this most clearly.

The $P_l^m(x)$ are not the only solutions to Legendre's associated equation. Just as there is a second set of solutions to Legendre's equation (5.77), so too there is a second set of solutions to Eq. (5.78). However, they also are not finite at $x = \pm 1$.

At this point we state an important result; the addition theorem for spherical harmonics. If two vectors $\mathbf{x}(r, \theta, \phi)$ and $\mathbf{x}'(r', \theta', \phi')$ have an angle γ between them, then

$$P_l(\cos\gamma) = \frac{4\pi}{2l+1} \sum_{m=-l}^{l} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) , \tag{5.83}$$

where

$$\cos\gamma = \frac{\mathbf{x}\cdot\mathbf{x}'}{|\mathbf{x}|\,|\mathbf{x}'|} = \cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\phi - \phi') . \tag{5.84}$$

Thus the addition theorem expresses a Legendre polynomial of order l in the angle γ in terms of a sum over products of spherical harmonics of the angles θ, ϕ and θ', ϕ'. We omit the proof; note, however, that for the case $l = 1$, Eq. (5.83) is just a statement of the familiar result of Eq. (5.84).

There is a raft of other properties of the spherical harmonics and associated Legendre functions that are straightforward, but messy, generalizations of the identities proved for Legendre polynomials. We shall not go into these further.

We devote the remainder of this section to a discussion of the spherical harmonics in their capacity as solutions to the angular part of Laplace's equation in spherical coordinates. Since the angular part of the Laplacian is essentially the quantum-mechanical operator that represents the total angular momentum squared, the spherical harmonics play a prominent role in quantum theory. The operator that represents the square of the total angular momentum is

$$L^2 = -\hbar^2\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right) . \tag{5.85}$$

Let us first see why this is the angular-momentum operator. Classically, $\mathbf{L} = \mathbf{r} \times \mathbf{p}$. If we make the usual Schrödinger substitution for $\mathbf{p}$: $\mathbf{p} \to i\hbar\nabla$, we

find that

$$
\begin{aligned}
L_x &= -i\hbar\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right), \\
L_y &= -i\hbar\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right), \\
L_z &= -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right).
\end{aligned}
\tag{5.86}
$$

Now we transform to spherical coordinates. The equations of transformation of coordinates are:

$$
\begin{aligned}
x &= r\sin\theta\cos\phi, & r &= (x^2 + y^2 + z^2)^{1/2}, \\
y &= r\sin\theta\sin\phi, & \phi &= \tan^{-1} y/x, \\
z &= r\cos\theta, & \theta &= \tan^{-1}\frac{(x^2 + y^2)^{1/2}}{z}.
\end{aligned}
$$

Therefore

$$
\begin{aligned}
\frac{\partial}{\partial x} &= \frac{\partial}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial x} + \frac{\partial}{\partial \phi}\frac{\partial \phi}{\partial x} \\
&= \sin\theta\cos\phi\frac{\partial}{\partial r} + \frac{\cos\theta\cos\phi}{r}\frac{\partial}{\partial \theta} - \frac{1}{r}\frac{\sin\phi}{\sin\theta}\frac{\partial}{\partial \phi}.
\end{aligned}
$$

Similarly,

$$
\begin{aligned}
\frac{\partial}{\partial y} &= \sin\theta\sin\phi\frac{\partial}{\partial r} + \frac{1}{r}\cos\theta\sin\phi\frac{\partial}{\partial \theta} + \frac{1}{r}\frac{\cos\phi}{\sin\theta}\frac{\partial}{\partial \phi}, \\
\frac{\partial}{\partial z} &= \cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial \theta}.
\end{aligned}
$$

Using these relations in Eqs. (5.86), we obtain after some manipulation,

$$
\begin{aligned}
L_x &= +i\hbar\left(\sin\phi\frac{\partial}{\partial \theta} + \cos\phi\cot\theta\frac{\partial}{\partial \phi}\right), \\
L_y &= -i\hbar\left(\cos\phi\frac{\partial}{\partial \theta} - \sin\phi\cot\theta\frac{\partial}{\partial \phi}\right), \\
L_z &= -i\hbar\frac{\partial}{\partial \phi}.
\end{aligned}
\tag{5.87}
$$

Now we form L^2:

$$
\begin{aligned}
L^2 = L_x^2 + L_y^2 + L_z^2 &= -\hbar^2\left[\frac{\partial^2}{\partial \theta^2} + \cot\theta\frac{\partial}{\partial \theta} + (1 + \cot^2\theta)\frac{\partial^2}{\partial \phi^2}\right] \\
&= -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial}{\partial \theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial \phi^2}\right].
\end{aligned}
\tag{5.88}
$$

We observe [see Eq. (1.68)] that the L^2 operator is essentially the angular part

of the Laplacian in spherical coordinates:

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{L^2}{\hbar^2 r^2} . \tag{5.89}$$

Thus we may write Laplace's equation in the form

$$\nabla^2\psi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) - \frac{L^2\psi}{\hbar^2 r^2} = 0 ,$$

or (5.90)

$$\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) = \frac{1}{\hbar^2}L^2\psi .$$

In order to solve this equation, let us assume that $\psi(r, \theta, \phi)$ is a product of two functions, one depending on r alone, and the other on θ and ϕ:

$$\psi(r, \theta, \phi) = R(r)\,Y(\theta, \phi) .$$

If we substitute a solution of this form into Eq. (5.90), and divide the equation by $\psi = RY$, we obtain

$$\frac{1}{R(r)}\frac{\partial}{\partial r}\left(r^2\frac{\partial R(r)}{\partial r}\right) = \frac{1}{\hbar^2 Y(\theta, \phi)}L^2 Y(\theta, \phi) .$$

The left-hand side is a function of r alone, and the right-hand side is a function only of θ and ϕ. This can only hold if both sides equal the same constant, for otherwise we could vary r and yet the right-hand side, depending only on θ and ϕ, would not change. Let us call the constant λ. Then we have the two equations

$$\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) = \lambda R(r) , \tag{5.91}$$

$$L^2 Y(\theta, \phi) = \hbar^2\lambda Y(\theta, \phi) . \tag{5.92}$$

It will be shown below that a solution for the angular equation (5.92) is just the spherical harmonic, Y_{lm} with $\lambda = l(l + 1)$. In other words, the spherical harmonics are eigenvectors of the operator L^2 corresponding to eigenvalues $\hbar^2 l(l + 1)$. For $\lambda = l(l + 1)$, the radial equation (5.91) may be solved by letting $R(r) = U(r)/r$. It then becomes

$$\frac{d^2}{dr^2}U(r) - l(l + 1)\frac{U(r)}{r^2} = 0 , \tag{5.93}$$

which has the solution

$$U(r) = rR(r) = Ar^{l+1} + Br^{-l} , \tag{5.94}$$

where A and B are constants. Together with the spherical harmonics, these radial functions give a solution to Laplace's equation in spherical coordinates. A function which satisfies Laplace's equation is called a harmonic function. This explains the origin of the name "spherical harmonics": these functions

are *harmonic*, since they satisfy the angular part of Laplace's equation, and they are defined on the surface of the unit *sphere*.

The most general solution is a linear combination of solutions for given values of l and m. Thus

$$\psi(r, \theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} [A_{lm} r^l + B_{lm} r^{-l-1}] Y_{lm}(\theta, \phi) . \tag{5.95}$$

It is of interest to separate the variables again, in Eq. (5.92). That is, let $Y_{lm} = \Phi(\phi) P(\theta)$, substitute into the equation, and divide through by $Y = \Phi P$. The result may be written

$$\frac{-\sin\theta}{P} \frac{d}{d\theta}\left(\sin\theta \frac{dP}{d\theta}\right) - l(l+1)\sin^2\theta = \frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} .$$

The θ- and ϕ-dependences have thus been isolated and so the expressions on both sides of the equation must equal the same constant, call it $-m^2$. We then have the two equations

$$\frac{d^2\Phi}{d\phi^2} = -m^2\Phi , \tag{5.96}$$

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right) + \left[l(l+1) - \frac{m^2}{\sin^2\theta}\right]P = 0 . \tag{5.97}$$

The azimuthal equation (in ϕ), Eq. (5.96), is immediately solved:

$$\Phi(\phi) = Ae^{\pm im\phi} . \tag{5.98}$$

For $\Phi(\phi)$ to be single-valued over the range $[0, 2\pi]$, m must be an integer. This result was anticipated in choosing the form of the separation constant. The equation in θ (Eq. 5.97) will assume a recognizable form if we simply change variables. Letting $\cos\theta = x$, so $dx = d(\cos\theta)$, and $\sin\theta = (1 - x^2)^{1/2}$, it becomes

$$(1 - x^2)P'' - 2xP' + \left[l(l+1) - \frac{m^2}{1 - x^2}\right]P = 0 , \tag{5.99}$$

which, due to our earlier choice of $\lambda = l(l+1)$, is just Legendre's associated equation (5.78). The solutions are the associated Legendre functions $P(x) = P_l^m(x)$ given in Eq. (5.81). This proves that the spherical harmonics are eigenvectors of L^2 with eigenvalues $\hbar^2 l(l+1)$, as stated above.

It should be noted that we really found *two* solutions to the second-order differential equation (5.91), $R(r) = r^l$ and $R(r) = r^{-l-1}$. This was also true for the differential equation (5.96), where we found the two solutions given by Eq. (5.98). However, for Eq. (5.99), which is also a second order differential equation, we have stated only one solution, $P_l^m(x)$. There is, of course, a second solution, just as there was to the other two equations. The second solution is rarely used in physical problems, however, because it is singular at $x = \pm 1$ ($\theta = 0, \pi$) and must usually be rejected in order to satisfy boundary conditions.

Usually one first encounters all these functions in their capacity as solutions to differential equations that arise in physics. Our approach has been different.

We have emphasized the *completeness* properties of these sets of functions, touching only incidentally on them as solutions to differential equations.

5.9 HERMITE POLYNOMIALS

In the next section we shall develop the properties of the various sets of orthogonal polynomials as special cases of a single comprehensive framework. But in order for us to be able to see the trees, and not just the forest, we shall first work out some of the properties of the Hermite polynomials in an inductive way. We first encountered these polynomials in the discussion of the eigenvalue problem for the quantum-mechanical harmonic oscillator (see Section 3.10).

The Hermite polynomials, $H_n(x)$, are the orthogonal polynomials on the infinite interval $[-\infty, \infty]$ with respect to the nonnegative weight function e^{-x^2}. That is, for $m \neq n$,

$$\int_{-\infty}^{\infty} H_n(x) H_m(x) e^{-x^2}\, dx = 0 . \tag{5.100}$$

Thus they differ in two important ways from the orthonormal sets of functions dealt with so far. First, they are defined from $-\infty$ to $+\infty$. Therefore Weierstrass' theorem, which deals only with finite intervals, does not apply. Secondly, they are orthogonal with respect to a weight function (see Definition 5.2). The Hermite polynomials are most conveniently *defined* in terms of a generating function $\psi(x, t)$:

$$\psi(x, t) \equiv e^{-t^2+2tx} = e^{x^2} e^{-(t-x)^2} \equiv \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n , \tag{5.101}$$

From this implicit definition of $H_n(x)$, we can get an explicit formula. It is clear that

$$\begin{aligned} H_n(x) &= \left[\frac{\partial^n \psi(x, t)}{\partial t^n}\right]_{t=0} = \left[e^{x^2} \frac{\partial^n}{\partial t^n} e^{-(t-x)^2}\right]_{t=0} \\ &= \left[e^{x^2} \frac{d^n}{dy^n} e^{-y^2}\right]_{y=-x} = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} . \end{aligned} \tag{5.102}$$

This is the Rodrigues formula for the Hermite polynomials. It shows that they are indeed polynomials of degree n.

We next derive some recursion relations for the Hermite polynomials. Since it is true of the generating function that

$$\frac{\partial \psi(x, t)}{\partial x} = 2t\, \psi(x, t) ,$$

it follows that

$$\sum_n \frac{H_n'(x) t^n}{n!} = \sum_n \frac{2H_n(x) t^{n+1}}{n!} .$$

Equating the coefficients of equal powers of t, we get

$$H_n'(x) = 2nH_{n-1}(x), \qquad n \geq 1 . \tag{5.103}$$

Likewise, from

$$\frac{\partial\psi(x,t)}{\partial t} + 2(t-x)\,\psi(x,t) = 0\,,$$

it follows that

$$H_{n+1}(x) - 2xH_n(x) + 2nH_{n-1}(x) = 0, \qquad n \geq 1\,. \tag{5.104}$$

The reader may show that, by differentiating these recursion relations and combining them properly, the following differential equation for the $H_n(x)$ can be derived:

$$H_n''(x) - 2x\,H_n'(x) + 2nH_n(x) = 0, \qquad n \geq 0\,. \tag{5.105}$$

There are many ways to prove orthogonality of the $H_n(x)$ with respect to the weight function e^{-x^2}. The two methods we have used for the Legendre polynomials—repeated integration by parts, and the proof based directly on the differential equation—also work for the Hermite polynomials. However, we shall use a different method, based on the generating function, in order to illustrate still another technique. Consider the integral

$$I = \int_{-\infty}^{\infty} e^{-t^2+2tx}e^{-s^2+2sx}e^{-x^2}\,dx = \sum_{n,m=0}^{\infty}\frac{t^n s^m}{n!m!}\int_{-\infty}^{\infty} H_nH_me^{-x^2}\,dx\,.$$

We want to prove that the integral on the right is zero for $m \neq n$. The integral on the left is

$$I = e^{-(t^2+s^2)}\int_{-\infty}^{\infty} e^{-x^2+2(s+t)x}\,dx = e^{-(t^2+s^2)}\pi^{1/2}e^{(s+t)^2} = \pi^{1/2}e^{2st}\,;$$

the integral is done by completing the square. Expanding this expression for I in a Taylor series, we have

$$I = \pi^{1/2}\sum_{n=0}^{\infty}\frac{(2st)^n}{n!} = \sum_{n,m=0}^{\infty}\frac{t^n s^m}{n!\,m!}\int_{-\infty}^{\infty} H_nH_m\,e^{-x^2}\,dx\,.$$

If equal powers of s and t are equated in this identity, we find that

$$\int_{-\infty}^{\infty} H_nH_me^{-x^2}\,dx = \pi^{1/2}2^n n!\delta_{nm}\,. \tag{5.106}$$

Thus the functions $u_n(x) = N_nH_n(x)e^{-x^2/2}$ with $N_n = (1/\pi^{1/2}2^n n!)^{1/2}$ are orthonormal on the interval $(-\infty, \infty)$. The functions $u_n(x)$ are the eigenfunctions of the quantum oscillator. We may say equivalently that the set of Hermite polynomials $H_n(x)$ are orthonormal on the interval $(-\infty, \infty)$ with respect to the weight function $N_n^2e^{-x^2}$.

Another important set of polynomials are the Laguerre polynomials, $L_n(x)$. They are orthogonal on the interval $[0, \infty]$ with respect to the weight function e^{-x}, that is

$$\int_0^{\infty} L_n(x)L_m(x)e^{-x}\,dx = 0 \qquad \text{for } n \neq m\,. \tag{5.107}$$

Everything that we have done for Hermite polynomials can be done for Laguerre polynomials (Problem 6.36).

5.10 STURM-LIOUVILLE SYSTEMS—ORTHOGONAL POLYNOMIALS

All the special functions we have studied so far are solutions to differential equations. These equations and their solutions look rather different at first, but, in fact, they have a great deal in common. In this section we shall view them as special cases in a single general framework.

All the equations are second-order, linear differential equations of the form

$$Lu = \lambda u \,, \tag{5.108}$$

where

$$L = \alpha(x)\frac{d^2}{dx^2} + \beta(x)\frac{d}{dx} + \gamma(x), \tag{5.109}$$

and λ is a constant; α, β, and γ are real functions of x. The operators L have an important property in common that until now we have not mentioned. Namely, they are *Hermitian* if (1) we use the inner product

$$(f, g) \equiv \int_{-\infty}^{\infty} f^*(x)g(x)w(x)\,dx \,, \tag{5.110}$$

[where the weight function $w(x) \geq 0$], and, (2) the functions u satisfy appropriate boundary conditions. We have not yet committed ourselves to a choice of $w(x)$, and we will find that this freedom is crucial in attaining the generality which we want.

In quantum mechanics, all the operators which correspond to physical observables are Hermitian. It will be recalled that a Hermitian operator H satisfies the equation

$$(Hx, y) = (x, Hy) \,,$$

where x and y are vectors.†

Now suppose that f and g are two vectors (functions) in Hilbert space. For the inner product we defined above, the condition that L be Hermitian is

$$(Lf, g) \equiv \int_{-\infty}^{\infty} (Lf(x))^* g(x)w(x)\,dx = \int_{-\infty}^{\infty} (f(x))^*(Lg(x))w(x)\,dx \equiv (f, Lg) \,. \tag{5.111}$$

Henceforth, we restrict attention to such operators.

To investigate the hermiticity of L, we compute (f, Lg) and (Lf, g) and then form their difference; L is Hermitian if and only if this difference is zero.

† Operators, H, satisfying $(Hx, y) = (x, Hy)$ on an infinite dimensional vector space are called *symmetric* operators by mathematicians. We will refer to them as Hermitian operators. They clearly have real eigenvalves and orthonormal eigenfunctions according to our work in Chapter 4.

We have

$$(f, Lg) = \int_{-\infty}^{\infty} wf^*\alpha g''\,dx + \int_{-\infty}^{\infty} wf^*\beta g'\,dx + \int_{-\infty}^{\infty} wf^*\gamma g\,dx\,.$$

Integrating the first two integrals by parts once, we get

$$\begin{aligned}(f, Lg) &= [w\alpha f^*g']_{-\infty}^{\infty} - \int_{-\infty}^{\infty} [(w\alpha)'f^*g' + (w\alpha)f^{*\prime}g']\,dx \\ &\quad + [w\beta f^*g]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} [(w\beta)'f^*g + (w\beta)f^{*\prime}g]\,dx + \int_{-\infty}^{\infty} wf^*\gamma g\,dx\,.\end{aligned}$$

We now subtract the corresponding expression for (Lf, g), which may be obtained from the above by simply interchanging f^* and g, and obtain

$$\begin{aligned}(f, Lg) - (Lf, g) &= [w\alpha(f^*g' - f^{*\prime}g)]_{-\infty}^{\infty} \\ &\quad - \int_{-\infty}^{\infty} [(w\alpha)' - (w\beta)](f^*g' - gf^{*\prime})\,dx\,.\end{aligned}$$

For L to be Hermitian, this last expression must be equal to zero for *all* f and g. Therefore the necessary and sufficient conditions for L to be Hermitian are

$$1. \qquad [w\alpha(f^*g' - gf^{*\prime})]_{-\infty}^{\infty} = 0\,, \tag{5.112}$$

$$2. \qquad (w\alpha)' = w\beta\,. \tag{5.113}$$

Clearly, the weight function $w(x)$ plays an important role in the Hermiticity requirement. In fact, the second equation above actually determines $w(x)$ up to a multiplicative constant, since $\alpha(x)$ and $\beta(x)$ are given functions. The first equation illustrates the importance of the endpoint conditions in determining Hermiticity. For example, if the functions on which L operates vanish sufficiently rapidly at the boundaries, then the first condition above is satisfied. The reader can easily show that there are several less stringent conditions which will fulfill condition one. On the other hand, if $w\alpha$ vanishes sufficiently rapidly at both endpoints, then the functions upon which L operates need not satisfy any special boundary conditions and condition (1) will still be satisfied. The requirement that $w\alpha$ vanish sufficiently rapidly at the endpoints can be met either by having $w\alpha$ identically zero outside some infinite interval $[a, b]$ or having $w\alpha$ fall off sufficiently rapidly at arbitrarily large distances. In what follows, we shall have occasion to utilize both possibilities.

Since $w(x)$ has not been specified, we may consider condition (2) as an equation to be solved for $w(x)$. We rewrite it as

$$(w\alpha)' = \frac{\beta}{\alpha} w\alpha\,. \tag{5.114}$$

A simple integration gives

$$w\alpha = C\cdot\exp\left[\int \frac{\beta}{\alpha}\,dx\right] \tag{5.115}$$

in any region in which $(w\alpha)'$ is continuous. The integral is an indefinite integral; C is an arbitrary constant. Note in particular that $C = 0$ $(w\alpha \equiv 0)$ is a solution. For future use, we should point out that it is possible to join a solution $w\alpha \not\equiv 0$ to a solution $w\alpha \equiv 0$ at some point $x = a$, if $w\alpha$ is continuous at this point, that is, $w\alpha = 0$ just to the right *and* the left of $x = a$ on the real line.

In the region where $w\alpha \not\equiv 0$, $\alpha(x)$ must either be nonnegative or nonpositive since we require $w(x) > 0$. In the former case, we must choose C positive, in the latter case, C must be negative. Without loss of generality, we can take $\alpha(x) \geq 0$ and $C > 0$. In summary, for any given $\alpha(x)$ and $\beta(x)$ there exists a weight function $w(x)$ given in any region where $w\alpha \not\equiv 0$ by Eq. (5.115). This guarantees that condition (2) is satisfied.

Any equation of the form we are studying,

$$Lu = \alpha(x)u'' + \beta(x)u' + \gamma(x)u = \lambda u\,, \tag{5.116}$$

can be written in the equivalent form

$$\frac{d}{dx}\left[w\alpha\frac{du}{dx}\right] + (\gamma - \lambda)wu = 0 \tag{5.117}$$

by choosing the weight function $w(x)$ according to Eq. (5.113). An equation of the form (5.117), together with the boundary condition stated in Eq. (5.112), is called a *Sturm-Liouville system*. In Eq. (5.46), Legendre's differential equation appears in Sturm-Liouville form. We used this form of Legendre's equation to prove the orthogonality of eigenfunctions belonging to distinct eigenvalues. We shall now give a general proof of this result, valid for any Sturm-Liouville system.

Let λ_m and λ_n $(\lambda_m \neq \lambda_n)$ be two eigenvalues of Eq. (5.117). Then

$$\frac{d}{dx}\left[(w\alpha)\frac{du_m}{dx}\right] + (\gamma - \lambda_m)wu_m = 0\,,$$

and

$$\frac{d}{dx}\left[(w\alpha)\frac{du_n^*}{dx}\right] + (\gamma - \lambda_n)wu_n^* = 0\,.$$

Multiplying the first of these equations by u_n^*, the second by u_m, and forming their difference, we obtain

$$u_n^*\frac{d}{dx}[(w\alpha)u_m'] - u_m\frac{d}{dx}[(w\alpha)u_n^{*\prime}] = -(\lambda_n - \lambda_m)wu_n^*u_m\,.$$

The left-hand side of this expression is a total derivative. Rewriting this, and integrating over both sides, we find that

$$[w\alpha(u_n^*u_m' - u_mu_n^{*\prime})]_{-\infty}^{\infty} = -(\lambda_n - \lambda_m)\int_{-\infty}^{\infty} wu_n^*u_m\,dx\,.$$

The left-hand side vanishes identically by assumption [Eq. (5.112)]; and since

$\lambda_n \neq \lambda_m$, it follows that

$$(u_m, u_n) = \int_{-\infty}^{\infty} u_m^*(x) u_n(x) w(x)\, dx = 0 .$$

Therefore eigenfunctions belonging to distinct eigenvalues are orthogonal with respect to the weight function $w(x)$. This result merely translates into function language the result on the orthogonality of the eigenvectors of Hermitian operators belonging to distinct eigenvalues (Theorem 4.18).

Let us now restrict our attention to *polynomial* Sturm-Liouville systems. That is, we shall assume that the functions $u_n(x)$ are polynomials of degree n. We denote them by $Q_n(x)$. Then

$$LQ_n = \lambda_n Q_n , \tag{5.118}$$

where

$$L = \alpha(x) \frac{d^2}{dx^2} + \beta(x) \frac{d}{dx} + \gamma(x) , \tag{5.119}$$

as before, and

$$\int_{-\infty}^{\infty} Q_n^*(x) Q_m(x) w(x)\, dx = 0 \tag{5.120}$$

if $m \neq n$, in keeping with the above discussion of orthogonality.

For this case, where Q_n is a polynomial of degree n, the real functions $\alpha(x)$, $\beta(x)$, and $\gamma(x)$ must have the form

$$\alpha(x) = \alpha_0 x^2 + \alpha_1 x + \alpha_2 , \tag{5.121}$$

$$\beta(x) = \beta_0 x + \beta_1 , \tag{5.122}$$

$$\gamma(x) = \gamma_0 . \tag{5.123}$$

That this is the most general form possible is a consequence of the fact that the differential equation is of second order and that we are considering only *polynomial* solutions. We prove this by letting $n = 0, 1$, and 2 in Eq. (5.118). Then

$$\gamma Q_0 = \lambda_0 Q_0 \Longrightarrow \gamma = \lambda_0 = \gamma_0 ,$$

$$\beta Q_1' + \gamma Q_1 = \lambda_1 Q_1 \Longrightarrow \beta = \frac{\lambda_1 - \lambda_0}{Q_1'} Q_1 = \beta_0 x + \beta_1 ,$$

$$\alpha Q_2'' + \beta Q_2' + \gamma Q_2 = \lambda_2 Q_2 \Longrightarrow \alpha = \frac{\lambda_2 - \lambda_0}{Q_2''} Q_2 - \frac{\lambda_1 - \lambda_0}{Q_1' Q_2''} Q_1 Q_2'$$
$$= \alpha_0 x^2 + \alpha_1 x + \alpha_2 .$$

We note that β can never be identically zero because if it were, then it follows from Eq. (5.113) that $w\alpha = \text{const}$. But this makes it impossible to satisfy the other requirement for Hermiticity (Eq. 5.112), since polynomials do not vanish at ∞.

Thus there are no polynomial solutions to the Sturm-Liouville problem (Hermitian L) if $\beta(x) \equiv 0$. If we do not insist on L being Hermitian, there are polynomial solutions of all degrees with eigenvalues equal to $n(n-1)\alpha_0$. The eigenvectors will not be orthogonal. The general expression for $\beta(x)$ above shows that $\lambda_1 = \lambda_0$ if $\beta(x) \equiv 0$. Conversely, if $\beta(x) \not\equiv 0$, then $\lambda_1 \neq \lambda_0$ and $\beta_0 \neq 0$ because Q_1 has a leading term proportional to x.

The restriction to polynomial solutions yields only a few of the possible Sturm-Liouville systems. For example, the well-known Bessel functions do not fall into this category. To ensure the Hermiticity of L, we shall satisfy Eq. (5.112) by choosing $w\alpha$ to vanish as $|x| \to \infty$. In fact, since in Eq. (5.112) $w\alpha$ is multiplied by f^*g' and $gf^{*\prime}$ which are now by assumption polynomials in x, we see that $w\alpha$ must actually tend to zero faster than any inverse power of x as $|x|$ tends to infinity. This also follows from the requirement that $(f, f) < \infty$ for all f in our space.

With these assumptions, all the properties of a polynomial Sturm-Liouville system now are determined by the six parameters α_0, α_1, α_2, β_0, β_1, and γ_0. However, there are four degrees of freedom for inessential changes in L:

1. L can be multiplied by a constant C_1. This leaves Q_n unchanged and multiplies the eigenvalues by C_1.
2. The independent variable x may be shifted by some constant C_2. Thus $Q_n(x)$ is replaced by $Q_n(x + C_2)$, and the eigenvalues are unchanged.
3. The independent variable x may be scaled by a constant C_3: $x \to C_3x$. Thus $Q_n(x)$ is replaced by $Q_n(C_3x)$, and the eigenvalues are left unchanged.
4. A constant may be added to L, that is, to γ_0. This leaves Q_n unchanged, and λ_n is replaced by $\lambda_n + C_4$. Throughout this section we shall, for convenience, choose C_4 so that $\gamma_0 \equiv 0$.

With γ_0 fixed (equal to zero), we have only five parameters left which characterize the Sturm-Liouville system (α_0, α_1, α_2, β_0, and β_1); but there are three inessential degrees of freedom remaining. Thus, in fact, the polynomial Sturm-Liouville system is completely characterized by $5 - 3 = 2$ parameters. We now consider the following three cases:

1. $\alpha(x)$ is quadratic;
2. $\alpha(x)$ is linear ($\alpha_0 = 0$), and
3. $\alpha(x)$ is a constant ($\alpha_0 = 0 = \alpha_1$).

Our discussion will completely exhaust the possible *polynomial* solutions to the second-order Sturm-Liouville differential equation. The reader will never be able to discover a new system of Sturm-Liouville polynomials and have them named after himself!

CASE 1. $\alpha(x)$ is quadratic. In this case, Eq. (5.115) is

$$w\alpha = C \cdot \exp\left[\int \frac{\beta_0 x + \beta_1}{\alpha_0 x^2 + \alpha_1 x + \alpha_2}\, dx\right], \tag{5.124}$$

where $\alpha_0 \neq 0$. There are two subcases to distinguish, depending on whether $\alpha(x)$ has real or complex roots. Let us consider the case of complex roots first, and for convenience pick C_1 so that $\alpha_0 = 1$. Then $\alpha(x) = (x - \kappa)(x - \kappa^*)$, where κ and κ^* are the two complex roots of $\alpha(x)$ (the asterisk, as usual, denotes complex conjugation). The integral in Eq. (5.124) is

$$w\alpha = C[\alpha(x)]^{\beta_0/2} \exp\left[\frac{\beta_1 + \beta_0 \operatorname{Re}\kappa}{|\operatorname{Im}\kappa|} \tan^{-1}\left(\frac{x - \operatorname{Re}\kappa}{|\operatorname{Im}\kappa|}\right)\right],$$

where $\operatorname{Re}\kappa$ and $\operatorname{Im}\kappa$ denote, respectively, the real and imaginary parts of κ. Since $\alpha(x)$ is a polynomial in x, $w\alpha$ cannot vanish faster than some inverse power of x as x tends to infinity (the precise power being determined by β_0), so it fails to satisfy our previously derived endpoint requirements, Eq. (5.112). Also, $w\alpha$ is never zero, so we cannot join it onto a solution $w\alpha \equiv 0$ outside some finite interval. Thus our only hope is to consider the case where $\alpha(x)$ has two real roots. For convenience in this case, let us choose the three inessential parameters so that $\alpha_0 = -1$ and the roots of $\alpha(x)$ are at $x = 1$ and $x = -1$. Thus $\alpha(x) = 1 - x^2$. Furthermore, let us adopt as a notational convenience the relations $\beta_1 = q - p$, $\beta_0 = -(p + q + 2)$; this particular convention is made so that $w(x)$ will work out in a nice form; p and q are real but need not be integers. Thus

$$\frac{\beta}{\alpha} = \frac{-(p + q + 2)x + q - p}{1 - x^2} = \frac{q + 1}{1 + x} - \frac{p + 1}{1 - x},$$

and by doing the simple integral in Eq. (5.115) we get

$$w\alpha = C(1 + x)^{q+1}(1 - x)^{p+1} \qquad \text{or} \qquad w(x) = C(1 + x)^q(1 - x)^p .$$

Once again it turns out that, at large distances, $w\alpha$ can tend to zero only as fast as some fixed inverse power of x (depending on the specific values of p and q), so in this form our endpoint requirements cannot be satisfied. However, if $q > -1$ and $p > -1$, $w\alpha$ vanishes at $x = 1$ and $x = -1$, so we can join the above expression for $w(x)$ on $[-1, 1]$ to the solution $w(x)$ identically equal to zero outside $[-1, 1]$. (Note that α_0 and β_0 always have the same sign, a fact we shall use subsequently.) Therefore all the objections we raised in the case of complex roots are overcome, and we have an acceptable weight function for the interval $[-1, 1]$:

$$w(x) = (1 + x)^q(1 - x)^p ,$$

where we have set $C = 1$.

The $Q_n(x)$ that correspond to these choices of the essential parameters for the case of $\alpha(x)$ quadratic are called *Jacobi polynomials* of index p, q, and they shall be designated $J_n^{(p,q)}(x)$. (The upper indices distinguish them from Bessel functions.) By making a linear change of the independent variable, they may be recast in another form. Letting $x \to 1 - 2x$ in the differential equation (5.119), we obtain

$$\alpha(x) = x(1 - x) \geq 0 \text{ in } [0, 1], \qquad \beta(x) = -(p + q + 2)x + (p + 1) .$$

Thus the left-hand endpoint is now $x = 0$; $\beta_1 = \beta(0) = p + 1$; and

$$\beta_0 = \beta(1) - \beta(0) = -(p + q + 2)$$

is unchanged. It follows that $\alpha w = x^{p+1}(1 - x)^{q+1}$, so $p, q > -1$, as before. The weight function is, therefore, now

$$w(x) = x^p(1 - x)^q .$$

The set of polynomials which are orthogonal to the interval $[-1, 1]$ with respect to the weight function $w(x) = (1 - x)^p(1 + x)^q$, and the polynomials orthogonal on $[0, 1]$ with respect to $w(x) = x^p(1 - x)^q$, differ only by a linear change of variable. Both are referred to as the Jacobi polynomials of index p, q.

We now deal with a number of special cases of the Jacobi polynomials as defined on $[-1, 1]$. For $p = q = m (m > -1)$, $w(x) = (1 - x^2)^m$. The corresponding polynomials are called *ultraspherical* or *Gegenbauer polynomials* of integral index m. We designate them by

$$G_n^m(x) = J_n^{(m, m)}(x) .$$

Note that $p = q$ implies $\beta_1 = 0$, which means that the operator L is unaltered by the change of variable $x \to -x$. If we call P the operator which changes x into $-x$, then the invariance of L under $x \to -x$ may be expressed in operator notation as $PLf(x) = LPf(x)$ for all $f(x)$, so

$$PL = LP ,$$

which according to Theorem (4.22) tells us that any eigenfunction of L can be chosen to be simultaneously an eigenfunction of P. Since $P^2f(x) = f(x)$ for all $f(x)$, we see that $P^2 = I$, the identity operator. This places severe limitations on the eigenvalues of P. Consider $Pf(x) = \lambda f(x)$. This implies

$$P^2f(x) = \lambda Pf(x) = \lambda^2 f(x) = f(x) ,$$

since $P^2 = I$. Thus $\lambda^2 = 1$, so $\lambda = \pm 1$. This means that any Gegenbauer polynomial can be characterized by a subscript $+$ or $-$, depending on whether it is an eigenfunction of P with eigenvalue $+1$ or -1 (see Problem 5.15).

P is called the *parity operator*. It plays a fundamental role in many areas of physics. According to the above discussion, one speaks of eigenvectors of positive or negative parity. Since the leading term in $G_n^m(x)$ is proportional to x^n and since $Px^n = (-1)^n x^n$, we see that the Gegenbauer polynomials have positive or negative parity depending on whether n is even or odd. If n is even, $G_n^m(x)$ contains *only* even powers of x, whereas if n is odd, $G_n^m(x)$ contains *only* odd powers of x. The Gegenbauer polynomials occur in the solution of certain relativistically invariant equations in the quantum theory of elementary particles.

If we consider the Jacobi polynomials with $p=q = -\frac{1}{2}$, then $w(x) = (1 - x^2)^{-1/2}$. We designate these polynomials

$$T_n(x) \equiv J_n^{(-1/2, -1/2)}(x) .$$

The orthogonality relation is

$$(T_{n_1}, T_{n_2}) = \int_{-1}^{1} T_{n_1}^*(x) T_{n_2}(x)(1 - x^2)^{-1/2}\, dx = 0\,, \qquad (n_1 \neq n_2).$$

If we set $x = \cos\theta$, we get the alternate expression

$$(T_{n_1}, T_{n_2}) = \int_{-\pi}^{\pi} T_{n_1}^*(\cos\theta) T_{n_2}^*(\cos\theta)\, d\theta = 0\,.$$

The only set of polynomials in $\cos\theta$ which obeys this relationship on $[-\pi, \pi]$ is

$$T_n(\cos\theta) \propto \cos n\theta\,.$$

Therefore in the x-variable on $[-1, 1]$,

$$T_n(x) \propto \cos(n \cos^{-1} x)\,,$$

so this set of polynomials is determined explicitly up to a constant factor. These polynomials are known as the *Tschebycheff polynomials*. It is probably correct to say that, until now, no two authors have ever spelled their name the same way.

The most familiar of all the polynomials, the Legendre polynomials, are sure to be found somewhere in this cascade of special cases. We see that they result if we take $p = q = m = 0$. Then $w(x) = 1$; thus $J_n^{(0,0)}(x) = P_n(x)$ up to constant factor.

There is a theorem (Problem 20) which tells us that if $\{Q_n\}$ is an orthonormal system of Sturm-Liouville polynomials, then the set $\{Q_n', n > 1\}$ is a system of orthogonal polynomials with weight function $w\alpha$. In the case of the Legendre polynomials, where $w = 1$ in $[-1, 1]$ and $\alpha = (1 - x^2)$, this result, applied μ times, tells us immediately that the set of polynomials $(d^\mu/dx^\mu)P_n$ is orthogonal on $[-1, 1]$ with respect to the weight function $\alpha^\mu = (1 - x^2)^\mu$. The uniqueness of sets of orthonormal polynomials on given intervals means that these polynomials are none other than the Gegenbauer polynomials:

$$G_{n-\mu}^{\mu} = \frac{d^\mu}{dx^\mu} P_n\,,$$

which is the only orthonormal set of polynomials with weight function $w = (1 - x^2)^\mu$ on $[-1, 1]$.

It also follows that the set of *functions*

$$P_n^\mu(x) = (1 - x^2)^{\mu/2} \frac{d^\mu}{dx^\mu} P_n(x)$$

is orthogonal with weight function 1 on $[-1, 1]$. These functions are the associated Legendre functions.

CASE 2. $\alpha(x)$ is a linear function. Under these circumstances, $\alpha(x) = x + \alpha_2$, where we have chosen C_1 so that $\alpha_1 = 1$. Also, $\alpha(x)$ has one real zero at $x = -\alpha_2$; adjusting our constant C_2, let us shift the zero to the origin, so finally

$\alpha(x) = x$. Thus

$$w\alpha = C\cdot\exp\left[\int \frac{\beta_0 x + \beta_1}{x}\,dx\right]$$
$$= C\cdot\exp\,(\beta_0 x + \beta_1 \ln x) = Cx^{\beta_1}e^{\beta_0 x}\,.$$

If we choose $\beta_0 < 0$, then $w\alpha \to 0$ as $x \to +\infty$, but $w\alpha$ grows exponentially as $x \to -\infty$. Thus to satisfy the boundary conditions, $w(x)$ must be chosen to vanish identically for all $x < a$, if possible. a is still to be determined. If we choose $\beta_1 > 0$, then $w\alpha = 0$ at $x = 0$ and the integral of $w\beta$ over an interval of length ϵ around $x = 0$ goes to zero as ϵ tends to zero, so we may correctly join our identically zero solution (for $x < 0$) to the above nontrivial solution at this point. By adjusting C_3 we may set $\beta_0 = -1$ for convenience. (Note that if we choose $\beta_0 > 0$, this would just reverse the situation, and $w\alpha$ would have to vanish on the positive x-axis. The choice $\beta_0 < 0$ is conventional.) Calling $\beta_1 = s + 1$, $s > -1$, we have

$$w(x) = x^s e^{-x}$$

for positive x and $w(x) = 0$ for negative x. The Q_n that correspond to these conditions are the *associated Laguerre polynomials* of order s. They are designated by $L_n^s(x)$. If $s = 0$, the functions $L_n^0(x) \equiv L(x)$ are called the *Laguerre polynomials*. The presence of only one free parameter (s) reflects the fact that $\alpha(x)$ is *linear* in x: $\alpha_0 = 0$, and since three of the four remaining coefficients in $\alpha(x)$ and $\beta(x)$ may be chosen arbitrarily, there is just a single free parameter left.

CASE 3. $\alpha(x)$ is a constant. Here $\alpha_0 = 0 = \alpha_1$, and by a judicious choice of C_1 we can arrange that $\alpha_2 = 1$. Thus

$$w\alpha = C\cdot\exp\left[\int (\beta_0 x + \beta_1)\,dx\right] = C\cdot\exp\left(\frac{\beta_0}{2}x^2 + \beta_1 x\right)$$
$$= C'\cdot\exp\left[\frac{\beta_0}{2}\left(x + \frac{\beta_1}{\beta_0}\right)^2\right].$$

If we choose $\beta_0 < 0$, then $w\alpha = w$ falls off exponentially as $|x| \to \infty$, so this expression for $w(x)$ is acceptable as it stands; choosing C_2 and C_3 appropriately, we can make $\beta_0 = -2$ and $\beta_1 = 0$, so setting $C' = 1$, we get

$$w(x) = e^{-x^2}\,.$$

The corresponding $Q_n(x)$ are the Hermite polynomials. We have now run through *all* the possibilities: there are no more Sturm-Liouville polynomials that are solutions of second-order differential equations. Table 5.1 summarizes the results obtained thus far.

We now work out the eigenvalues belonging to each of the three basic cases. We need only collect the terms in x^n in the differential equation. This gives

$$\lambda_n = \alpha_0 n(n-1) + \beta_0 n = n(\alpha_0 n + \beta_0 - \alpha_0)\,. \tag{5.125}$$

It follows immediately that the eigenvalues are spaced linearly for the Hermite

and Laguerre equations (for which $\alpha_0 = 0$), and quadratically for Jacobi's equation, and all the special cases of it.

In particular, for Hermite's equation we have $\lambda_n = -2n$, and for Laguerre's equation we have $\lambda_n = -n$. For Jacobi's equation the eigenvalues are $\lambda_n = -n(n + p + q + 1)$ regardless of whether the equation is the one whose solutions are orthogonal on $[0, 1]$ or the equation whose solutions are orthogonal on $[-1, 1]$. The Gegenbauer special case ($p = q = m$) has eigenvalues $\lambda_n = -n(n + 2m + 1)$. The two further specialized cases, Tschebycheff's ($m = -\frac{1}{2}$) and Legendre's ($m = 0$) equations, have eigenvalues $\lambda_n = -n^2$ and $\lambda_n = -n(n + 1)$, respectively. It should be clear that whether the eigenvalues are positive or negative is merely a matter of the specific sign conventions adopted elsewhere in the treatment.

We conclude this discussion of polynomial Sturm-Liouville systems by proving a general Rodrigues formula which provides explicit expressions for all the polynomials up to a constant factor. The general formula is

$$Q_n(x) = K_n \frac{1}{w} \frac{d^n}{dx^n} (\alpha^n w) , \tag{5.126}$$

where the choice of the constant K_n depends on the physical application. The proof we shall give is a general version of the one in Section 5.5 for the Legendre polynomials. Any set of polynomials is uniquely determined by the requirement that $Q_n(x)$ is a polynomial of nth degree that satisfies the orthogonality relation

$$\int_a^b Q_m^* Q_n w \, dx = 0 , \qquad \text{for } n \neq m . \tag{5.127}$$

Thus we only have to show that the $Q_n(x)$, as given by the generalized Rodrigues formula [Eq. (5.126)], are polynomials of degree n, and satisfy Eq. (5.127) and we shall have proved Eq. (5.126).

We first prove a preliminary result: If $f(x) = \alpha^k w r(x)$, where $r(x)$ is a polynomial of degree l, then $f'(x) = \alpha^{k-1} w s_{l+1}(x)$, where $s_{l+1}(x)$ is a polynomial of degree $l + 1$. To prove this we simply differentiate $f(x)$:

$$\begin{aligned} f'(x) &= (\alpha^{k-1})'(\alpha w) r + \alpha^{k-1}(\alpha w)' r + \alpha^{k-1}(\alpha w) r' \\ &= (k-1)\alpha^{k-2}\alpha'(\alpha w) r + \alpha^{k-1}(\beta w) r + \alpha^{k-1}(\alpha w) r' \\ &= \alpha^{k-1} w[\beta r + (k-1)\alpha' r + \alpha r'] \\ &= \alpha^{k-1} w[r_0(\beta_0 + (2k + l - 2)\alpha_0) x^{l+1} + \cdots] \\ &= \alpha^{k-1} w s_{l+1}(x) . \end{aligned}$$

We have written $r(x) = r_0 x^l + \cdots$. Since $\beta_0 \neq 0$, and since α_0 and β_0 have the same sign if $\alpha_0 \neq 0$, then the expression in brackets is a polynomial of degree $l + 1$ if $k \geq 1$ which is the only situation of interest. From this result applied n times to $\alpha^n w$, ($r = 1$), we infer that

$$\frac{d^n}{dx^n} (\alpha^n w) = \alpha^0 w t(x) = w t_n(x) ,$$

Table 5.1

THE STURM-LIOUVILLE SYSTEMS OF ORTHOGONAL POLYNOMIALS

Name of polynomials	Differential equation	α_0	α_1	α_2	β_0	β_1	$w\alpha$	w	$[a, b]$
(I) *α quadratic*									
Jacobi	$(1-x^2)u'' + [-(p+q+2)x + (q-p)]u' + n(n+p+q+1)u = 0$	-1	0	1	$-(p+q+2)$	$q-p$	$(1-x)^{p+1}(1+x)^{q+1}$	$(1-x)^p(1+x)^q$	$[-1, 1]$
Jacobi	$(x-x)u'' + [-(p+q+2)x + (p+1)]u' + n(n+p+q+1)u = 0$	-1	1	0	$-(p+q+2)$	$p+1$	$x^{p+1}(1-x)^{q+1}$	$x^p(1-x)^q$	$[0, 1]$
Gegenbauer	$(1-x^2)u'' - 2(m+1)xu' + n(n+2m+1)u = 0$	-1	0	1	$-2(m+1)$	0	$(1-x^2)^{m+1}$	$(1-x^2)^m$	$[-1, 1]$
Tschebycheff	$(1-x^2)u'' - xu' + n^2u = 0$	-1	0	1	-1	0	$(1-x^2)^{1/2}$	$(1-x^2)^{-1/2}$	$[-1, 1]$
Legendre	$(1-x^2)u'' - 2xu' + n(n+1)u = 0$	-1	0	1	-2	0	$1-x^2$	1	$[-1, 1]$
(II) *α linear*									
Laguerre	$xu'' - [x-(s+1)]u' + nu = 0$	0	1	0	-1	$s+1$	$x^{s+1}e^{-x}$	x^se^{-x}	$[0, \infty]$
(III) *α* constant									
Hermite	$u'' - 2xu' + 2nu = 0$	0	0	1	-2	0	e^{-x^2}	e^{-x^2}	$[-\infty, \infty]$

where $t(x)$ is a polynomial of degree n. It follows from Eq. (5.126) that $Q_n(x)$ is indeed a polynomial of degree n.

We can now prove the orthogonality of the $Q_n(x)$ with respect to $w(x)$. We have

$$\begin{aligned}\int_{-\infty}^{\infty} Q_m^*(x)Q_n(x)w(x)\,dx &= \int_{-\infty}^{\infty} Q_m^*(x)\,\frac{d^n}{dx^n}\,(\alpha^n w)\,dx \\ &= \sum_{k=0}^{n-1}\left[(-1)^k\left(\frac{d^k}{dx^k}\,Q_m^*\right)\frac{d^{n-1-k}}{dx^{n-1-k}}\,(\alpha^n w)\right]_{-\infty}^{\infty} \\ &\quad + (-1)^n\int_{-\infty}^{\infty}\left(\frac{d^n}{dx^n}\,Q_m^*\right)\alpha^n w\,dx\,. \qquad (5.128)\end{aligned}$$

Here we have integrated by parts n times. If $n > m$, the integral vanishes, so orthogonality will be satisfied if we can show that the terms $d^l/dx^l(\alpha^n w)$, $0 \leq l \leq n-1$, all vanish at the endpoints faster than any inverse power of x. But this follows from the preliminary result which tells us that

$$\frac{d^l}{dx^l}\,(\alpha^n w) = \alpha^{n-l} w v_l(x)\,,$$

where $v_l(x)$ is a polynomial of degree l. Since by assumption αw vanishes at the endpoints faster than any inverse power of x, so will $\alpha^{n-l} w v_l(x)$ for all $0 \leq l \leq n-1$, and the proof of Eq. (5.126) is complete.

In carrying out this proof we have shown that, in general,

$$\int_{-\infty}^{\infty} R(x)Q_n(x)w(x)\,dx = (-1)^n\int_{-\infty}^{\infty}\frac{d^n R}{dx^n}\,\alpha^n w\,dx\,. \qquad (5.129)$$

This relation can be useful in the evaluation of integrals. In particular, if we take $R(x) \equiv Q_n(x)$, the normalization integral is

$$\int_{-\infty}^{\infty} Q_n^2(x)w(x)\,dx = (-1)^n n!\,q_n\int_{-\infty}^{\infty}\alpha^n w\,dx\,,$$

where q_n is the leading coefficient of $Q_n(x)$, as given in Eq. (5.126).

We can now use this generalized Rodrigues formula to obtain expressions for the polynomials we have considered. The Rodrigues formula will only provide us with expressions up to constant factors, which may then be selected in several ways. In some books they are selected in such a way as to make the coefficient of the highest power in $Q_n(x)$ equal to one for all n. Sometimes they are chosen so as to normalize the polynomials with respect to the corresponding weight function; and sometimes in still other ways. For example, the Legendre polynomials defined in Eq. (5.38) are not normalized on $[-1, 1]$. The normalized Legendre polynomials are given by $(n+\frac{1}{2})^{1/2}\,P_n(x)$. One must beware of the multiple inconsistent conventions that exist in the literature in defining the various sets of orthogonal polynomials. We shall not put in explicit expressions for the constant factors in the list that follows, except for the Legendre and Hermite polynomials, which we have worked out earlier.

Now, using the generalized Rodrigues formula (Eq. 5.126), we obtain for the Jacobi polynomials,

$$J_n^{(p,q)}(x) = A_n^{(p,q)}(1-x)^{-p}(1+x)^{-q}\frac{d^n}{dx^n}\left[(1-x^2)^n(1-x)^p(1+x)^q\right], \quad (5.130)$$

where $A_n^{(p,q)}$ is a constant factor depending on p, q, and n.

For the Gegenbauer polynomials, $p = q = m$, so this becomes

$$G_n^m(x) = B_n^m(1-x^2)^{-m}\frac{d^n}{dx^n}(1-x^2)^{n+m}, \quad (5.131)$$

where B_n^m is a constant factor depending on m and n.

For the Tschebycheff ($m = -\frac{1}{2}$) and Legendre ($m = 0$) polynomials, we have

$$T_n(x) = C_n(1-x^2)^{1/2}\frac{d^n}{dx^n}(1-x^2)^{n-1/2}, \quad (5.132)$$

where C_n is a constant factor depending on n;

$$P_n(x) = \frac{(-1)^n}{2^n n!}\frac{d^n}{dx^n}(1-x^2)^n. \quad (5.133)$$

For the associated Laguerre polynomials, we obtain

$$L_n^s(x) = D_n^s x^{-s}e^x\frac{d^n}{dx^n}(x^{n+s}e^{-x}), \quad (5.134)$$

where D_n^s is a constant factor which may depend on s and n; conventions vary widely in its choice. The Laguerre polynomials are the case $s = 0$:

$$L_n(x) = D_n e^x\frac{d^n}{dx^n}(n^x e^{-x}); \quad (5.135)$$

here D_n depends only on n.

Finally, the Hermite polynomials are given by

$$H_n(x) = (-1)^n e^{x^2}\frac{d^n}{dx^n}e^{-x^2}. \quad (5.136)$$

We close this section by showing that all the Sturm-Liouville polynomials form complete sets in Hilbert space when the inner product in Hilbert space is given by

$$(f, g) \equiv \int_{-\infty}^{\infty} f^*(x)g(x)w(x)\,dx,$$

with $w(x)$ given for the various polynomials in Table 5.1. In this discussion we shall take into account the Hermite and Laguerre polynomials which are defined on infinite intervals and therefore do not fall within the scope of Weierstrass's Theorem. The proof of completeness is based on Theorem 5.2 which says that if a set of orthonormal functions is closed, then it is complete.

Completeness Theorem. The orthonormal set of Sturm-Liouville polynomials $\{Q_n(x)\}$ is complete in Hilbert space.

Proof. We shall show that the set $\{Q_n(x)\}$ is closed. Suppose there is some function $f(x)$ orthogonal to all the $Q_n(x)$, so that $(f, Q_n) = 0$ for all n. Since x^m can be written as a finite linear combination of the $Q_n(x)$, the first $m + 1$ to be exact, we see immediately that $(x^m, f) = 0$ for all m. Now consider

$$g(k) \equiv (e^{ikx}, f) = \int_{-\infty}^{\infty} f(x)w(x)e^{-ikx}\,dx\,,$$

for all real k. Because of the presence of $w(x)$ in the inner product, e^{-ikx} will belong to Hilbert space for any $w(x)$ appropriate to a Sturm-Liouville polynomial; so will x^m. Now, if we expand e^{-ikx} as

$$e^{-ikx} = \sum_{m=0}^{\infty} \frac{(-ik)^m}{m!} x^m$$

and use $(x^m, f) = 0$, we see that $g(k) = 0$ for all k. But $g(k)$ is precisely the Fourier transform of $f(x)\ w(x)$. Hence by our discussion in Section 5.7, $f(x)\ w(x) = 0$ almost everywhere. Thus we conclude that on any interval where $w(x) \neq 0$, $f(x) = 0$ almost everywhere. We are only interested in $f(x)$ on intervals where $w(x) \neq 0$, since we can define $f(x)$ to be anything we like where $w(x) = 0$. Another way to put this is to say that the Hilbert space is really defined only on the interval where $w(x) \neq 0$, since when $w(x) = 0$ on some finite interval, the inner product gives no structure there at all.

In summary, we conclude that $f(x) = 0$ almost everywhere, so the set $\{Q_n(x)\}$ is closed, and therefore it is a complete orthonormal set. QED

These sets of orthogonal polynomials have many additional polynomials. For example, using the generalized Rodrigues formula, one can derive recursion relations for the various sets of polynomials. Also, the zeros of the polynomials can be characterized in some detail. The interested reader is referred to Lyusternik or Wilf. In Section 6.10 we shall go on to establish the equivalence of the explicit Rodrigues representation of some of these polynomials and their implicit representation in terms of a generating function.

All these sets of functions are solutions to differential equations, as are the common trigonometric functions. All of them are related by various identities, just as the sine and cosine functions are. It is useful to think about them in this light. We study sines and cosines first, but these particular functions are not essentially different from Legendre or Hermite polynomials, or any of the other special functions of physics. They are more familiar to us only because we can apply them to the measurement of distances and because they turn up in macroscopic simple harmonic motion. If we imagine very small intelligent beings who do not measure large distances and instead first investigate quantum effects, it is reasonable to suppose that they discover Hermite polynomials at an early date by investigating the quantum oscillator. After centuries of work,

they spend half their Gross National Product to build a (to us) macroscopic oscillator (to them it will be absolutely gigantic), and, in the course of this project, they discover the trigonometric functions. To them, sines and cosines will seem complicated and troublesome compared to the old familiar Hermite polynomials.

5.11 A MATHEMATICAL FORMULATION OF QUANTUM MECHANICS

In this and in the preceding chapters we have developed many of the mathematical ideas used in the formulation of quantum mechanics, namely, the concepts of vector spaces and, in particular, Hilbert space. We shall now write and interpret the axioms of quantum mechanics in terms of these mathematical tools. What we give here is not a rigorous axiomatization, but rather an introduction to a profound and beautiful subject. For more complete treatments, the reader should see J. M. Jauch's book, *Foundations of Quantum Mechanics*, or J. Von Neumann's classic work, *Mathematical Foundations of Quantum Mechanics*.

The mathematical setting for quantum mechanics is Hilbert space, that is, a complete complex inner-product space. As we have seen, the elements of this space are functions. To give a mathematical formulation of quantum mechanics, we must also make use of *operators* on Hilbert space. In our discussion of the properties of these operators, we will rely heavily on the results of Chapters 3 and 4. We have already noted that the validity of many of the theorems in Chapters 3 and 4 is not restricted to finite-dimensional spaces, and we shall make use of these theorems where they are applicable. We shall also use them to arrive intuitively at certain important generalizations of finite-dimensional results which will be central to our treatment.

We begin by discussing the notion of a *projection* operator (Problem 4.17), which will prove very useful in the mathematical formulation of quantum mechanics. We do this in the context of some familiar finite-dimensional results. Consider an Hermitian operator, A, whose eigenvalues and eigenvectors are $\{\lambda_n\}$ and $\{\phi_n\}$ respectively ($n = 1, 2, \cdots, N$). We assume, for notational convenience, that λ_n is nondegenerate. We now define the operator I by

$$I\psi \equiv \sum_{n=1}^{N} \phi_n(\phi_n, \psi)$$

for any ψ in the finite-dimensional space in question. According to the completeness theorem (Theorem 4.20), this sum is equal to ψ, so that the operator defined in this way is in fact the identity operator, as we had anticipated by our choice of notation. If we define P_n by

$$P_n\psi \equiv \phi_n(\phi_n, \psi) \ , \tag{5.137}$$

then I can be written as

$$I = \sum_{n=1}^{N} P_n \ .$$

The reader can easily verify that $P_n^2 = P_n$, $P_n^\dagger = P_n$, and $P_n P_m = 0$ if $m \neq n$. P_n is a *projection operator*, so called because it projects any vector into the one-dimensional subspace of V spanned by ϕ_n.

What can we say about the operator, B, defined by

$$B \equiv \sum_{n=1}^{N} \lambda_n P_n \, ?$$

Since

$$B\phi = \sum_{n=1}^{N} \lambda_n \phi_n (\phi_n, \phi) \ ,$$

it is immediately clear that B has eigenvalues $\{\lambda_n\}$ with corresponding eigenvectors $\{\phi_n\}$. It must therefore be equal to A (the reader will find it instructive to prove this). In summary, we have

$$I = \sum_{n=1}^{N} P_n \ , \qquad A = \sum_{n=1}^{N} \lambda_n P_n \ . \tag{5.138}$$

Thus we may restate the completeness theorem as follows: For every self-adjoint operator, A, there exists a set of orthogonal projections, $\{P_n\}$, such that Eqs. (5.138) are satisfied. By *orthogonal* projections, we mean projections which satisfy $P_n P_m = 0$ for $m \neq n$. It is in this language that one can generalize the completeness theorem to infinite-dimensional spaces. However, it turns out that only for a special type of infinite-dimensional operator (the completely continuous type, which we will discuss in Chapters 8 and 9) do Eqs. (5.138) generalize directly.

In an infinite-dimensional space, there arises the possibility of "eigenvalues" which are infinitely closely spaced, so that the above formulation needs some alteration. To get the flavor of what these alterations involve, we must say a word about an important generalization of the Riemann integral—the Stieltjes integral. We define

$$\int_a^b f(x)\, dg(x) \equiv \lim \sum_{i=1}^{N} f(\bar{x}_i)[g(x_{i+1}) - g(x_i)] \ , \tag{5.139}$$

where the points $x_1, x_2, \cdots, x_{n+1}$ represent a partition of the interval $[a, b]$, and $\bar{x}_i$ is any point in the ith interval. "Lim" denotes the passage to very small partition intervals. Obviously, if $g(x)$ is a nice differentiable function, we may write

$$\int_a^b f(x)\, dg(x) = \int_a^b f(x) \frac{dg}{dx}\, dx \ ,$$

and we get back to the familiar Riemann integral. However, $g(x)$ need not be differentiable or even continuous. Consider the function

$$g(x) = \begin{cases} 1 & \text{if} \quad x \geq 0 \\ 0 & \text{if} \quad x < 0 \ . \end{cases}$$

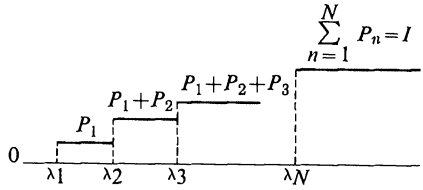

Fig. 5.5. A schematic plot of $E(\lambda)$ for the special case of a finite-dimensional self-adjoint operator whose eigenvalues are $\lambda_1, \lambda_2, \cdots, \lambda_N$.

According to the definition of Eq. (5.139), we have simply

$$\int_{-\infty}^{\infty} f(x)\, dg(x) = f(0) ,$$

since only an arbitrarily small interval around $x = 0$ can contribute to the sum in Eq. (5.139). This result is a δ-function type equation, which is not surprising since in the imprecise δ-function language,

$$\frac{d}{dx} g(x) = \delta(x) .$$

However, in the Stieltjes treatment, we never go beyond the domain of very traditional mathematics.

Now let us define an operator-valued function $E(\lambda)$ as follows:

$$E(\lambda) = \begin{cases} 0 & \text{for} \quad \lambda < \lambda_1 , \\ \sum_{n=1}^{\nu} P_n & \text{for} \quad \lambda_\nu \leqslant \lambda < \lambda_{\nu+1} , \qquad \nu = 1, \cdots, N-1 , \\ \sum_{n=1}^{N} P_n & \text{for} \quad \lambda \geq \lambda_N ; \end{cases}$$

the P_n and λ_n are defined above. This operator-valued function is "plotted" schematically in Fig. 5.5. Note that

$$E(-\infty) = 0 \qquad \text{and} \qquad E(+\infty) = I.$$

The reader can easily verify that

$$E(\lambda_1) E(\lambda_2) = E(\bar{\lambda}),$$

where $\bar{\lambda}$ is the smaller of λ_1 and λ_2 (this follows from the fact that the P_n are *orthogonal* projections), and that

$$[E(\lambda)]^2 = E(\lambda).$$

Using the definition of the Stieltjes integral (Eq. 5.139), we see that

$$\int_{-\infty}^{\lambda} dE(\lambda) = \sum_{n=1}^{\nu} P_n$$

if λ lies between λ_ν and $\lambda_{\nu+1}$. The only contributions to this integral come from the points at which $E(\lambda)$ changes discontinuously; at each such point, λ_i (which must be an eigenvalue of A by construction), we pick up the projection operator onto the one-dimensional subspace of V which is spanned by ϕ_i. In this new way of writing things, the first of Eqs. (5.138) becomes

$$I = \int_{-\infty}^{\infty} dE(\lambda) \ ,$$

and the second of Eqs. (5.138) can be written as

$$A = \int_{-\infty}^{\infty} \lambda dE(\lambda) \ .$$

Thus we have succeeded in making the simple finite-dimensional completeness theorem look extremely complicated.

Now let us look at the virtues of this complexity. Consider the familiar problem

$$\frac{d^2\phi_n(x)}{dx^2} = \lambda_n \phi_n(x) \ .$$

We know that with periodic boundary conditions on $[-L, L]$

$$\phi_n = \frac{1}{\sqrt{2L}} e^{i(n\pi/L)x} \ , \qquad \lambda_n = \frac{n^2\pi^2}{L^2} \ , \qquad n = 0, \pm 1, \pm 2, \cdots$$

As we let L become very large, the eigenvalues cluster very close together, so that in Fig. 5.5 the intervals of constancy of $E(\lambda)$ become increasingly small. We may imagine that in this limit $E(\lambda)$ tends to a *continuous* operator-valued function. It is this phenomenon which is given the name of "continuous spectrum," and it is very familiar in quantum mechanics—indeed, it may be said to lie at the heart of quantum mechanics.

Having introduced the function $E(\lambda)$ on a finite-dimensional space, we will generalize, without proof, to an infinite-dimensional Hilbert space. The following theorem is one of the most powerful results obtained in mathematics and is the key to a rigorous formulation of quantum mechanics.

Spectral Theorem. To every self-adjoint operator, A, on a Hilbert space, H, there corresponds an operator-valued function (unique), $E(\lambda)$, such that

1. $E(\lambda_1)E(\lambda_2) = E(\bar{\lambda})$; $\bar{\lambda} = \min(\lambda_1, \lambda_2)$,
2. $\lim_{\lambda\to-\infty} E(\lambda) = 0$, $\lim_{\lambda\to\infty} E(\lambda) = I$,
3. $I = \int_{-\infty}^{\infty} dE(\lambda)$,
4. $A = \int_{-\infty}^{\infty} \lambda dE(\lambda)$.

$E(\lambda)$ is called a *resolution of the identity* belonging to A. The set of points at which $E(\lambda)$ is nonconstant is referred to as the *spectrum* of A. For all λ, $E(\lambda)$ commutes with A and with any transformation which commutes with A.

Obviously, the spectral theorem reduces to the familiar completeness theorem in a finite-dimensional space. There are several remarks which should be made at this point. First, note that we have not been precise in saying what we mean by the limit of a sequence of operators (see Part 2 of the above theorem). We shall say more on this point in Chapters 8 and 9, but it is worth remarking that in nonrelativistic quantum mechanics, the spectrum is always bounded below (i.e., we cannot have bound states with arbitrarily large negative energy) so that the limit $\lambda \to -\infty$ is unambiguous. Second, we remark that if A is an *unbounded* operator (i.e., if there are elements of H on which A is *not defined*) then self-adjoint does not mean merely $(Af, g) = (f, Ag)$ whenever Af and Ag are defined. We will not dwell on this point; suffice it to say that all interesting operators which occur in nonrelativistic quantum mechanics are self-adjoint.

With these mathematical preliminaries out of the way, we now proceed to discuss the axioms of quantum mechanics.

Axiom I. Any physical system is completely described by a normalized vector ψ (the *state vector* or *wave function*) in Hilbert space. All possible information about the system can be derived from this state vector by rules which will be given in the following axioms.

Axiom II. To every physical observable there corresponds a self-adjoint operator on Hilbert space.

Examples of observables are position, momentum, energy, angular momentum, spin, etc. This axiom brings with it the powerful machinery associated with the spectral theorem. Notice that Axiom II does not specify how one is to discover the self-adjoint operator which corresponds to a given observable. It merely says that one exists. In practice, the use of classical expressions such as $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, $E = p^2/2m + V(\mathbf{r})$, etc., along with the operator substitutions $\mathbf{p} \to -i\hbar\nabla$ and $\mathbf{r} \to \mathbf{r}$ has been very successful. However, in the case of electron spin, one is forced beyond classical analogy to find the required operator.

Axiom III. The only allowed physical results of measurements of the observable A are the elements of the spectrum of the operator which corresponds to A.

We have already discussed some examples in which the spectrum is discrete. For instance, the operator L^2 of Section 5.8, which corresponds to the

square of the total angular momentum of a particle, satisfies

$$L^2 Y_{lm} = \hbar^2 l(l+1) Y_{lm} .$$

Here there are $2l + 1$ eigenvectors corresponding to each eigenvalue, $\hbar^2 l(l+1)$. Another system whose spectrum is particularly simple is the quantum harmonic oscillator. The classical Hamiltonian of this system is

$$\mathscr{H} = p^2/2m + \tfrac{1}{2}kx^2,$$

and by making the operator substitutions suggested above, we find

$$H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}kx^2$$

for the *operator* H corresponding to the energy. We saw in Sections 3.10 and 4.4 that this operator has eigenvalues $\lambda_n = \hbar\omega_0(n + \frac{1}{2})$, where $\omega_0 = (k/m)^{1/2}$, and normalized eigenvectors

$$\phi_n(x) = (\sqrt{\pi}\, 2^n n!)^{-1/2} H_n(\alpha x) e^{-\alpha^2 x^2/2} ,$$

where

$$\alpha = (m\omega_0/\hbar)^{1/2} ,$$

and H_n is the nth Hermite polynomial. We have already seen in this chapter that this set of ϕ_n is a complete orthonormal set in Hilbert space.

Now let us look at the situation for a very simple quantum-mechanical system whose spectrum is strictly continuous. The operator corresponding to the momentum of a particle is $p_x = -i\hbar\, d/dx$ (we consider just the x-component). In the space of square-integrable functions (such functions must tend to zero at large distances), p_x is self-adjoint, so according to the spectral theorem, we expect that we should be able to find a resolution of the identity belonging to p_x. If we were very free and easy about this problem, we would simply say that p_x had "eigenfunctions" $\phi_k = (2\pi)^{-1/2} e^{ikx}$ and "eigenvalues" $\hbar k$ for all k. The factor $(2\pi)^{-1/2}$ in ϕ_k is chosen to "normalize e^{ikx} to a δ-function," that is,

$$\int_{-\infty}^{\infty} \phi_k^*(x)\phi_{k'}(x)\, dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i(k'-k)x}\, dx = \delta(k - k') .$$

Also,

$$\int_{-\infty}^{\infty} \phi_k^*(x)\phi_k(x')\, dk = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{ik(x'-x)} = \delta(x - x') ,$$

which one often thinks of as being analogous to

$$\sum_{n=1}^{N} P_n = I$$

in finite-dimensional spaces. That is, we imagine a P_k defined by

$$P_k f = \phi_k(\phi_k, f) ,$$

where k is a continuous variable $(-\infty < k < \infty)$. Note, however, that P_k is not a conventional operator, since it pushes vectors *out of Hilbert space*. This is due to the fact that ϕ_k is not an element of Hilbert space, and hence is not an eigenvector of p_x in the customary sense. However, this approach does suggest what a correct solution of the problem would be. On the basis of our earlier work, we expect that if we want to obtain a resolution of the identity, $E(k)$, then this resolution should be the sum of all the projection operators P_k, with $k' < k$. Thus we might try

$$E(k)f = \int_{-\infty}^{k} \phi_{k'}(\phi_{k'}, f)\, dk' \ .$$

Written out in detail, this is

$$E(k)f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{k} e^{ik'x} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ik'y} f(y)\, dy \right] dk' \ ,$$

or

$$E(k)f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{k} e^{ik'x} \hat{f}(k')\, dk' \ , \tag{5.140}$$

where $\hat{f}(k')$ is the Fourier transform of $f(x)$. We have already indicated that the Fourier transform of every element in the space of square-integrable functions exists and is itself square-integrable (we will prove this rigorously in Chapter 9). Since $\hat{f}(k')$ is square-integrable, the integral in Eq. (5.140) clearly exists. It is a simple exercise to show that the vector $E(k)f(x)$ lies in Hilbert space.

We see by inspection that $E(\infty) = I$, by our main result on the inversion of Fourier transforms, and since $\hat{f}(k')$ is square-integrable, $E(-\infty) = 0$. Now we must show that $E(k_1)E(k_2) = E(k_1)$ (where we assume without loss of generality that $k_1 < k_2$). We reason as follows:

$$\begin{aligned} E(k_1)E(k_2)f &= E(k_1) \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{k_2} e^{ik'x} \hat{f}(k')\, dk' \\ &= \frac{1}{2\pi} \int_{-\infty}^{k_1} e^{ikx} F(k)\, dk \ , \end{aligned}$$

where $F(k)$ denotes the Fourier transform of

$$\int_{-\infty}^{k_2} e^{ik'x} \hat{f}(k')\, dk'$$

considered as a function of x. This transform is readily computed:

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} \int_{-\infty}^{k_2} e^{ik'x} \hat{f}(k') dk'\, dx = \begin{cases} \sqrt{2\pi}\, \hat{f}(k) & \text{if} \quad k_2 > k \ , \\ 0 & \text{if} \quad k_2 < k \ . \end{cases}$$

This result can be most simply obtained by the use of Eq. (5.63). Thus, since $k_1 < k_2$,

$$E(k_1)E(k_2)f = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{k_1} e^{ikx} \hat{f}(k)\, dk = E(k_1)f \ ,$$

as required. Finally, the reader can verify for himself that

$$\int_{-\infty}^{\infty} dE(k) = I ,$$

so we have indeed found a resolution of the identity.

To complete our task, we need only show that the above resolution belongs to p_x, that is, that

$$p_x = \hbar \int_{-\infty}^{\infty} k \, dE(k) .$$

This is easily done. We write

$$\begin{aligned}
\left[\hbar \int_{-\infty}^{\infty} k \, dE(k)\right] f &= \frac{\hbar}{\sqrt{2\pi}} \int_{-\infty}^{\infty} k \, d\left[\int_{-\infty}^{k} e^{ik'x} \hat{f}(k') \, dk'\right] \\
&= \frac{\hbar}{\sqrt{2\pi}} \int_{-\infty}^{\infty} k \frac{d}{dk}\left[\int_{-\infty}^{k} e^{ik'x} \hat{f}(k') \, dk'\right] dk \\
&= \frac{\hbar}{\sqrt{2\pi}} \int_{-\infty}^{\infty} k \, e^{ikx} \hat{f}(k) \, dk \\
&= \frac{-i\hbar}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{d}{dx} e^{ikx} \hat{f}(k) \, dk \\
&= \frac{-i\hbar}{\sqrt{2\pi}} \frac{d}{dx} \int_{-\infty}^{\infty} e^{ikx} \hat{f}(k) \, dk = -i\hbar \frac{df}{dx} = p_x f
\end{aligned}$$

for any f for which df/dx lies in our Hilbert space (under these conditions the above interchange of integration and differentiation is permissible). Thus we have found the resolution of the identity belonging to p_x. Of course, the spectral theorem tells us that such a resolution must exist; we have now explicitly constructed it. As we expect, the spectrum of p_x is purely continuous, consisting of all real numbers. Using the informal ideas above, the reader should be able to show that the resolution of the identity to the position operator X, defined by $Xf(x) \equiv xf(x)$, is the operator-valued function $E(\xi)$, specified by

$$E(\xi)f(x) = \begin{cases} f(x) & \text{if } x \leqslant \xi , \\ 0 & \text{if } x > \xi . \end{cases}$$

For more complicated self-adjoint operators, it is more difficult to find an explicit resolution of I, but it is always possible in principle.

We are now in a position to state the central axiom of quantum mechanics.

Axiom IV. If, on a system in the state ψ, we make a measurement of the observable A, then the probability that the value λ obtained by this measurement will lie between λ_1 and λ_2 ($\lambda_2 > \lambda_1$) is given by

$$P(\lambda_1, \lambda_2) = ||[E(\lambda_2) - E(\lambda_1)]\psi||^2 ,$$

when $E(\lambda)$ is the resolution of the identity belonging to A (by Axiom II plus the Spectral theorem).

Since $||\psi|| = 1$, we see that the probability of finding λ between $-\infty$ and $+\infty$ is 1, as it must be if the theory is to make sense physically. If $E(\lambda)$ is constant between λ_1 and λ_2 (i.e., if there are no points of the spectrum of A between λ_1 and λ_2), then $E(\lambda_2) - E(\lambda_1) = 0$, so a value can never be found in this range. This is in accord with Axiom III; in fact, it is clear that Axiom III is contained in Axiom IV. We have put in Axiom III only because its statement is such a familiar part of elementary quantum mechanics; if we were trying for minimal redundancy, we would omit Axiom III.

A particularly important case is that in which $E(\lambda)$ is constant between λ_1 and λ_2 *except* for a discontinuous jump at $\bar{\lambda}$. We will look at this case in some detail both for its importance to physics and because it enables us to show how a discrete spectrum can emerge from the projection-operator formalism. Intuitively, we should expect from our original motivations that whenever we have a discontinuity in $E(\lambda)$ at, say, $\bar{\lambda}$, on each side of which are intervals of constancy of $E(\lambda)$, we should find an eigenfunction of A. This is easily shown to be the case. Let λ_+ be any point in the interval of constancy to the right of $\bar{\lambda}$, and let λ_- be any point in the interval of constancy to the left of $\bar{\lambda}$. By analogy with our starting point in finite-dimensional spaces, we expect that if ψ is any element of H such that $\phi_{\bar{\lambda}} \equiv [E(\lambda_+) - E(\lambda_-)]\psi \neq 0$, then $\phi_{\bar{\lambda}}$ is an eigenfunction of A. We argue as follows:

$$\begin{aligned} A[E(\lambda_+) - E(\lambda_-)]\psi &= \int_{-\infty}^{\infty} \lambda\, d\{E(\lambda)[E(\lambda_+) - E(\lambda_-)]\psi\} \\ &= \int_{-\infty}^{\infty} \lambda\, d[E(\lambda)E(\lambda_+)\psi] - \int_{-\infty}^{\infty} \lambda\, d[E(\lambda)E(\lambda_-)\psi] \\ &= \int_{-\infty}^{\lambda_+} \lambda\, d[E(\lambda)\psi] - \int_{-\infty}^{\lambda_-} \lambda\, d[E(\lambda)\psi] \\ &= \int_{\lambda_-}^{\lambda_+} \lambda\, d[E(\lambda)\psi]\,. \end{aligned}$$

But by our assumption of intervals of constancy on either side of $\bar{\lambda}$, we have

$$\int_{\lambda_-}^{\lambda_+} \lambda\, d[E(\lambda)\psi] = \bar{\lambda}[E(\lambda_+) - E(\lambda_-)]\psi = \bar{\lambda}\phi_{\bar{\lambda}}\,.$$

Hence

$$A\phi_{\bar{\lambda}} = \bar{\lambda}\phi_{\bar{\lambda}}\,,$$

so $\bar{\lambda}$ is an eigenvalue of A, and $\phi_{\bar{\lambda}}$ is an eigenvector belonging to that eigenvalue.

Of course, $\phi_{\bar{\lambda}}$ may not be the *only* eigenvector belonging to $\bar{\lambda}$. If one picked a different ψ to start with (note that it is always possible to find a ψ such that $[E(\lambda_+) - E(\lambda_-)]\psi \neq 0$), one might find a vector $\tilde{\phi}_{\bar{\lambda}}$ which belongs to $\bar{\lambda}$ but is not just a multiple of $\phi_{\bar{\lambda}}$. In this case, we say that $\bar{\lambda}$ has multiplicity $\mu > 1$. We leave it as an exercise for the reader to show that if $\Phi_{\bar{\lambda}}$ is the unique normalized eigenvector belonging to $\bar{\lambda}$ (that is, $\bar{\lambda}$ has multiplicity 1), then for any $\psi \in H$,

$$[E(\lambda_+) - E(\lambda_-)]\psi = \Phi_{\bar{\lambda}}(\Phi_{\bar{\lambda}}, \psi)\,, \tag{5.141}$$

where λ_+ and λ_- are as chosen above. Thus, just as we would expect from our original finite-dimensional analogy, the discontinuity in $E(\lambda)$ at an eigenvalue $\bar{\lambda}$ is just the projection operator onto the one-dimensional subspace of H spanned by $\Phi_{\bar{\lambda}}$. If $\bar{\lambda}$ has multiplicity μ greater than one, then the discontinuity in $E(\lambda)$ at $\bar{\lambda}$ is just the projection operator onto the μ-dimensional subspace of H spanned by the μ eigenvectors belonging to $\bar{\lambda}$. Thus, in general, we may write

$$[E(\lambda_+) - E(\lambda_-)]\phi = \sum_{\nu=1}^{\mu} \Phi_{\bar{\lambda}}^{(\nu)} (\Phi_{\bar{\lambda}}^{(\nu)}, \phi) \ , \tag{5.142}$$

where μ is the multiplicity of $\bar{\lambda}$ (see Problem 5.25).

Returning now to our interpretation of Axiom IV, we see that in the case of a discontinuous jump in $E(\lambda)$ between λ_1 and λ_2, the probability of obtaining a measured value of A between λ_1 and λ_2 is, according to Eq. (5.141), just

$$P(\lambda_1, \lambda_2) = |(\Phi_{\bar{\lambda}}, \phi)|^2 \ ,$$

where $\bar{\lambda}$ is the point of discontinuity; we have assumed that $\bar{\lambda}$ has multiplicity 1. Thus $P(\lambda_1, \lambda_2)$ is just the modulus squared of the component of ϕ along the "direction" in Hilbert space defined by the eigenvector $\Phi_{\bar{\lambda}}$ belonging to $\bar{\lambda}$. More generally, if $\bar{\lambda}$ has multiplicity μ,

$$P(\lambda_1, \lambda_2) = \sum_{\nu=1}^{\mu} |(\Phi_{\bar{\lambda}}^{(\nu)}, \phi)|^2 \ . \tag{5.143}$$

Hence Eqs. (5.141) and (5.142) lead to very simple results for the probability of finding an observable A in the discrete spectrum.

If we choose an observable with a continuous spectrum, the situation is similar. Consider the x-component of the momentum, p_x. By our previous work, the probability of finding p_x between $\hbar k_1$ and $\hbar k_2$ is just

$$\begin{aligned} P(k_1, k_2) &= ||[E(k_2) - E(k_1)]\phi|| \\ &= \left\| \frac{1}{\sqrt{2\pi}} \int_{k_1}^{k_2} e^{ikx} \hat{\phi}(k)\, dk \right\| \\ &= \int_{k_1}^{k_2} |\hat{\phi}(k)|^2\, dk \ , \end{aligned}$$

where we have again used Eq. (5.63). In a more conventional language, $P(k_1, k_2)$ is just the continuous superposition of "plane-wave" probability amplitudes.

We now define the *expectation value* of A.

Definition. $\langle A \rangle \equiv \lim \sum_i \bar{\lambda}_i ||[E(\lambda_i + \Delta) - E(\lambda_i)]\phi||^2$ is the *expectation value* of the observable A for the system described by the state vector ϕ. Here we have divided the spectrum into intervals of length Δ, and $\bar{\lambda}_i$ is any point in the ith interval. By lim we mean the limit as the interval size tends to zero.

The physical meaning of the expectation value is very simple. It is just the value that would be found by taking the average of many measurements of

A on a large collection of systems all in the state ψ. Each possible result, $\bar{\lambda}_i$, of a measurement is weighted by the probability of finding a result in the vicinity of $\bar{\lambda}_i$. Note that $\langle A \rangle$ is a number, not an operator.

Theorem. $\langle A \rangle = (\psi, A\psi)$.

Proof. According to the definition (Eq. 5.139) of the Stieltjes integral,

$$\begin{aligned}(\psi, A\psi) &= \int_{-\infty}^{\infty} \lambda \, d[(\psi, E(\lambda)\psi)] \\ &= \lim \sum_i \bar{\lambda}_i (\psi, [E(\lambda_i + \Delta) - E(\lambda_i)]\psi) \ .\end{aligned}$$

But $[E(\lambda_i + \Delta) - E(\lambda_i)]^2 = [E(\lambda_i + \Delta) - E(\lambda_i)]$, so using the self-adjointness of E, we have

$$\begin{aligned}(\psi, A\psi) &= \lim \sum_i \bar{\lambda}_i ([E(\lambda_i + \Delta) - E(\lambda_i)]\psi, [E(\lambda_i + \Delta) - E(\lambda_i)]\psi) \\ &= \lim \sum_i \bar{\lambda}_i ||(E(\lambda_i + \Delta) - E(\lambda_i)]\psi||^2 = \langle A \rangle\end{aligned}$$

by definition.

In terms of the expectation value of A, we define the mean-square deviation $(\Delta A)^2$, which measures the dispersion around the mean value $\langle A \rangle$.

Definition. $(\Delta A)^2$ is the expectation value of $[A - \langle A \rangle]^2$ in the state ψ in which $\langle A \rangle$ is computed.

Theorem. $(\Delta A)^2 = \langle A^2 \rangle - \langle A \rangle^2$.

Proof.

$$\begin{aligned}(\Delta A)^2 &= \langle [A - \langle A \rangle]^2 \rangle = (\psi, [A - \langle A \rangle]^2 \psi) \\ &= (\psi, [A^2 - 2A\langle A \rangle + \langle A \rangle^2]\psi) \\ &= (\psi, A^2\psi) - 2(\psi, A\psi)\langle A \rangle + \langle A \rangle^2 \\ &= \langle A^2 \rangle - \langle A \rangle^2 \ ,\end{aligned}$$

as required. We have used the previous theorem several times in obtaining this result.

Note that according to our definition,

$$(\Delta A)^2 = (\psi, [A - \langle A \rangle]^2 \psi) = ([A - \langle A \rangle]\psi, [A - \langle A \rangle]\psi) \ ,$$

because A is self-adjoint and $\langle A \rangle$ is real. Thus

$$(\Delta A)^2 = ||[A - \langle A \rangle]\psi||^2 \ ,$$

so if $\Delta A = 0$ for some ψ, we must have $[A - \langle A \rangle]\psi = 0$; that is, ψ is an eigenvector of A with eigenvalue $\langle A \rangle$. This has the important implication that unless ψ is an eigenstate of A, one cannot make a perfectly precise measurement of A in that state. Thus only values of A lying in the discrete spectrum of A can be determined with perfect accuracy. For other values, one must always have some dispersion, ΔA, associated with measurements.

It is instructive to check that if $\psi = \Phi_n$, a normalized eigenstate of A with eigenvalue λ_n, then indeed $\Delta A = 0$. We have

$$\langle A \rangle = (\Phi_n, A\Phi_n) = \lambda_n ,$$
$$\langle A^2 \rangle = (\Phi_n, A^2\Phi_n) = (A\Phi_n, A\Phi_n) = \lambda_n^2 ,$$

so

$$\langle \Delta A \rangle^2 = \langle A^2 \rangle - \langle A \rangle^2 = \lambda_n^2 - \lambda_n^2 = 0 .$$

Furthermore, if $\psi = \Phi_n$, the probability of obtaining a value λ between λ_1 and λ_2 when A is measured is zero if $[\lambda_1, \lambda_2]$ does not contain λ_n, and this probability is 1 if $[\lambda_1, \lambda_2]$ contains λ_n. This follows simply by showing that the probability of finding λ in a small interval around λ_n is 1. According to Axiom IV,

$$\begin{aligned} P(\lambda_n - \epsilon, \lambda_n + \epsilon) &= ||[E(\lambda_n+\epsilon) - E(\lambda_n - \epsilon)]\Phi_n||^2 \\ &= |(\Phi_n, \Phi_n)|^2 = 1 , \end{aligned}$$

where we have made use of Eq. (5.142). Thus the probability of the measured value of A being λ_n is 1. In other words, if a system is in an eigenstate, so that its state vector happens to be an eigenvector of A and not a superposition of many eigenvectors, then the result of a measurement of A is certain to be the eigenvalue corresponding to that eigenvector.

Having considered in detail the case of a single measurement, let us mention briefly an important generalization of Axiom IV which admits the possibility of simultaneous measurement of several observables.

Axiom IV′. Let A, B, and C be observables whose corresponding linear operators commute, that is, $[A, B] = [A, C] = [B, C] = 0$. Then the probability that a simultaneous measurement of A, B, and C in a system whose state vector is ψ will yield a value of A between a_1 and a_2, B between b^1 and b_2, and C between c_1 and c_2 is

$$\begin{aligned} P(a_1, a_2; b_1, b_2; c_1, c_2) = ||&[E_A(a_2) - E_A(a_1)][E_B(b_2) - E_B(b_1)] \\ &\times [E_C(c_2) - E_C(c_1)]\psi||^2 , \end{aligned}$$

where $E_A(a)$, $E_B(b)$, and $E_C(c)$ are the resolutions of the identity belonging to A, B, and C, respectively.

In the case of a single measurement, Axiom IV′ is clearly equivalent to Axiom IV.

Note that since A, B, and C commute, so do $E_A(a)$, $E_B(b)$, and $E_C(c)$, and therefore the ordering of the projection operators on the right-hand side of the above equation is immaterial. Clearly, if their order were significant, the concept of simultaneous measurement would be untenable. The possibility of simultaneous measurement of *noncommuting* observables is denied in quantum mechanics; thus p_x and x, which satisfy $[x, p_x] = i\hbar$, cannot be measured simultaneously. The importance of simultaneous measurement of observables will become apparent when we discuss Axiom VI below.

Everything said so far has been concerned with the properties of a system at a fixed time. We now must say what happens to the system as the time changes.

Axiom V. For every system, there exists an Hermitian operator, H, the Hamiltonian or energy operator, which determines the time development of the state vector, Ψ, of the system through the time-dependent Schrödinger equation:

$$H\Psi(x, t) = i\hbar \frac{\partial \Psi(x, t)}{\partial t} , \tag{5.144}$$

provided that the system is not disturbed (by a measurement, for example). Here $\hbar$ is a constant equal to 1.054×10^{-27} erg-sec.

Axiom I says that the state vector Ψ is normalized. To be certain that our axioms are consistent, we must therefore show that this normalization is preserved in time by the above equation which governs the time dependence of the state vector. Thus we prove the following result:

Theorem.

$$\frac{\partial}{\partial t}(\Psi, \Psi) = 0 .$$

Proof.

$$\frac{\partial}{\partial t}(\Psi, \Psi) = \left(\frac{\partial \Psi}{\partial t}, \Psi\right) + \left(\Psi, \frac{\partial \Psi}{\partial t}\right) .$$

But

$$\frac{\partial \Psi}{\partial t} = \frac{1}{i\hbar} H\Psi ,$$

so,

$$\begin{aligned} \frac{\partial}{\partial t}(\Psi, \Psi) &= \left(\frac{1}{i\hbar} H\Psi, \Psi\right) + \left(\Psi, \frac{1}{i\hbar} H\Psi\right) \\ &= -\frac{1}{i\hbar}(H\Psi, \Psi) + (\Psi, H\Psi)\frac{1}{i\hbar} \\ &= -\frac{i}{i\hbar}(\Psi, H\Psi) + (\Psi, H\Psi)\frac{1}{i\hbar} = 0 , \end{aligned}$$

since H is self-adjoint. Thus Axiom I is consistent with Axiom V.

In general, the solution of Schrödinger's time-dependent equation for a Hamiltonian with complicated time dependence is a difficult job. However, if the Hamiltonian is time-independent, then matters simplify considerably. In this case, it is elementary to verify that the solution to Eq. (5.144), which reduces to $\Psi(x, 0)$ at $t = 0$, is simply

$$\Psi(x, t) = e^{-iHt/\hbar}\Psi(x, 0) . \tag{5.145}$$

Since H is self-adjoint, we have a resolution of the identity belonging to H, and therefore Eq. (5.145) can be written as

$$\Psi(x, t) = \int_{-\infty}^{\infty} e^{-i\epsilon t/\hbar} d[E(\epsilon)\Psi(x, 0)] \ . \tag{5.146}$$

Again, it is elementary to verify directly that this $\Psi(x, t)$ satisfies Eq. (5.144). Note that if the spectrum of H is purely discrete, then according to Eq. (5.143), Eq. (5.146) takes on the simple form

$$\Psi(x, t) = \sum_n e^{-i\epsilon_n t/\hbar}(\phi_n, \Psi(t = 0))\phi_n(x) \ , \tag{5.147}$$

where the ϕ_n are the eigenvectors of H and the ϵ_n are its eigenvalues. According to Axiom IV, the probability of finding a value ϵ_n for the energy of the system whose state vector is $\Psi(x, 0)$ is just

$$P(\epsilon_n;\ t = 0) = |(\phi_n, \Psi(t = 0))|^2 \ .$$

Using Eq. (5.147), we see that at time t, the probability of measuring H and finding ϵ_n is

$$P(\epsilon_n;\ t) = |e^{-i\epsilon_n t/\hbar}(\phi_n, \Psi(t = 0)|^2 = |(\phi_n, \Psi(t = 0))|^2 = P(\epsilon_n;\ t = 0);$$

that is, the probability of finding a particular value of the energy does not change with time.

Let us now see what happens for more general observables. Suppose that A is any observable, with associated resolution of the identity, $E_A(\lambda)$. According to Axiom IV, the probability of finding a value of A between λ_1 and λ_2 is just

$$P(\lambda_1, \lambda_2;\ t) = ||[E_A(\lambda_2) - E_A(\lambda_1)]e^{-iHt/\hbar}\Psi(x, 0)||^2 \ ,$$

where we have made use of Eq. (5.145) and are continuing to assume that H is independent of time. It is clear that, in general, $P(\lambda_1, \lambda_2;\ t)$ will depend on time, but there are two important situations in which $P(\lambda_1, \lambda_2;\ t)$ will be time-independent.

1. $\Psi(x, 0) = \phi_n(x)$, where $\phi_n(x)$ is an eigenvector of H.

   In this case,

$$P(\lambda_1, \lambda_2;\ t) = ||E_A(\lambda_2) - E_A(\lambda_1)]e^{-i\epsilon_n t/\hbar}\phi_n(x)||^2 \ ,$$

   but since $e^{-i\epsilon_n t/\hbar}$ is just a complex number with modulus 1, it cannot affect the norm of a vector, so

$$P(\lambda_1, \lambda_2;\ t) = ||[E_A(\lambda_2) - E_A(\lambda_1)]\phi_n||^2 = P(\lambda_1, \lambda_2;\ t = 0) \ .$$

2. H commutes with A.

   In this case, H also commutes with $E_A(\lambda)$, so

$$P(\lambda_1, \lambda_2;\ t) = ||e^{-iHt/\hbar}[E_A(\lambda_2) - E_A(\lambda_1)]\Psi(x, 0)||^2 \ .$$

But according to Theorem 4.13, this is simply

$$P(\lambda_1, \lambda_2; t) = ||[E_A(\lambda_2) - E_A(\lambda_1)]\Psi(x,0)||^2 = P(\lambda_1, \lambda_2; t = 0) ,$$

since $\exp[-iHt/\hbar]$ is a unitary operator. This tells us that the results of measuring an observable which commutes with the Hamiltonian are independent of the time at which they are measured.

By way of illustration, let us now investigate the situation for a particular observable, the position of a particle. We have already introduced the operator X defined by $Xf(x) \equiv xf(x)$ and have stated (see Problem 5.26) that its resolution of the identity is given by

$$E_X(\xi)f(x) = \begin{cases} f(x) & \text{if} \quad x \leq \xi , \\ 0 & \text{if} \quad x > \xi . \end{cases}$$

We now ask, what is the probability of obtaining a value of the position between ξ_1 and ξ_2 for a particle in a state Ψ? According to Axiom IV,

$$P(\xi_1, \xi_2; t) = ||[E_X(\xi_2) - E_X(\xi_1)]\Psi(x, t)||^2 .$$

From the definition of $E_X(\xi)$, we see that the vector whose norm is to be evaluated is just the function which is zero outside $[\xi_1, \xi_2]$ and equal to $\Psi(x, t)$ inside $[\xi_1, \xi_2]$. Thus

$$P(\xi_1, \xi_2; t) = \int_{\xi_1}^{\xi_2} |\Psi(x, t)|^2 dx .$$

Note that the probability of finding the particle *someplace* is 1, since Ψ is normalized. If we set $\xi_1 = \xi - \Delta/2$ and $\xi_2 = \xi + \Delta/2$, then for Δ small, we see that the probability of finding the particle in a small interval of length Δ about ξ is just $P = |\Psi(\xi, t)|^2\Delta$; thus $|\Psi(x, t)|^2$ is called the *probability density*, $\rho(x, t)$. This result is often stated at the beginning of quantum mechanics courses. It arose first, historically, in the pioneering work of Born on the probabilistic interpretation of quantum mechanics. Here the result is presented as a special case of Axiom IV. Note that, in general, the probability of finding the particle in a particular region varies with time; however, if $\Psi(x, t)$ is an eigenstate of the Hamiltonian, then $\Psi(x, t)$ is related to $\Psi(x, 0)$ just by a multiplicative phase factor, so $|\Psi(x, t)|^2 = |\Psi(x, 0)|^2$, and the probability of finding the particle in any region is constant in time. This is why eigenstates of the Hamiltonian are often called *stationary states*.

Finally, let us use Axiom V to obtain $P(\xi_1, \xi_2; t)$ in terms of a particular initial state, $\Psi(x, t = 0)$. For simplicity, we assume that $\Psi(x, 0)$ is given by

$$\Psi(x, 0) = (1/\sqrt[4]{\pi a^2})e^{-x^2/2a^2} .$$

Thus the initial probability density is equal to $(1/\pi^{1/2}a)e^{-x^2/a^2}$, so that the probability of finding the particle at a distance greater than a from the origin is very small compared with the probability of finding the particle within a distance a on either side of the origin. Let us assume that we are dealing with a free particle (and confine our attention to the x-dependence of the problem). The